

Neuronal migration depends on blood flow in the adult brain

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This **fundamental** work provides novel insights into the blood flow-dependent mechanisms of neuronal migration and the role of Ghrelin signaling in the adult brain. The authors present **convincing** evidence that newborn rostral migratory stream (RMS) neurons are closely situated alongside blood vessels, preferentially along arterioles, and that migratory speed is correlated with blood flow. They also provide evidence (in vitro and some in vivo) that Ghrelin from blood is involved in augmenting RMS neuron migration speed.

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Abstract

In animal tissues, several cell types migrate along blood vessels, raising the possibility that blood flow influences cell migration. Here, we show that blood flow promotes the migration of new olfactory-bulb neurons in the adult brain. Neuronal migration is facilitated by blood flow, leading to accumulation of new neurons near blood vessels with abundant blood flow. Blood flow inhibition attenuates blood vessel-guided neuronal migration, suggesting that blood contains factors beneficial to neuronal migration. We found that ghrelin, which is increased in blood by hunger, directly influences neuronal migration. Ghrelin signaling promotes somal translocation by activating actin cytoskeleton contraction at the rear of the cell soma. New neurons mature in the olfactory bulb and contribute to the olfactory function for sensing odorants from food. Finally, we show that neuronal migration is increased by calorie restriction, and that ghrelin signaling is involved in the process. This study suggests that blood flow promotes neuronal migration through blood-derived ghrelin signaling in the

adult brain, which could be one of the mechanisms that improves the olfactory function for food-seeking behavior during starvation.

Introduction

Blood vessel-guided cell migration has been reported in various types of cells, including lymphatic endothelial cells in the embryonic zebrafish trunk (Bussmann et al., 2010 [DOI](#)), Schwann cells, oligodendrocyte precursor cells, glioma cells, and astrocyte progenitors in rodent tissues (Cattin et al., 2015 [DOI](#); Farin et al., 2006 [DOI](#); Tabata et al., 2022 [DOI](#); Tsai et al., 2016 [DOI](#)), suggesting that it is a common mechanism for efficient cell migration within complex animal tissues. Previous reports have extensively studied the role of blood vessels as a physical scaffold for migration of neurons produced in the postnatal brain tissues (Bovetti et al., 2007 [DOI](#); Fujioka et al., 2017 [DOI](#); Grade et al., 2013 [DOI](#); Ohab et al., 2006 [DOI](#); Snapyan et al., 2009 [DOI](#); Sun et al., 2015 [DOI](#); Thored et al., 2007 [DOI](#); Whitman et al., 2009 [DOI](#); Yamashita et al., 2006 [DOI](#)). New neurons generated in the postnatal ventricular-subventricular zone (V-SVZ) migrate to the olfactory bulb (OB) through the rostral migratory stream (RMS), forming elongated aggregates called chains (Doetsch and Alvarez-Buylla, 1996 [DOI](#); Lois et al., 1996 [DOI](#); Lois and Alvarez-Buylla, 1994 [DOI](#)). Within the OB, new neurons migrate individually and radially and terminate their migration in the granule cell layer (GCL) or glomerular layer (GL), where they are differentiated into GABAergic interneurons and incorporated into the OB neural circuit (Bressan and Saghatelian, 2021 [DOI](#); Kaneko et al., 2017 [DOI](#)). Previous studies have shown that both chain-forming and individually migrating neurons use blood vessels as physical scaffolds for their migration in the RMS and OB (Bovetti et al., 2007 [DOI](#); Snapyan et al., 2009 [DOI](#); Whitman et al., 2009 [DOI](#)). However, whether blood flow plays a role in blood vessel-guided cell migration has not been reported.

Ghrelin, a peripheral hormone produced in the stomach, is delivered to the brain through the bloodstream and accumulates in the brain parenchyma, especially at high levels in the OB (Rhea et al., 2018 [DOI](#)). The ghrelin concentration in blood has been reported to increase during fasting (Toshinai et al., 2001 [DOI](#); Tschöp et al., 2000 [DOI](#)). In addition to its role in regulating metabolism (Al Massadi et al., 2017 [DOI](#); Chopin et al., 2012 [DOI](#); Soleyman-Jahi et al., 2019 [DOI](#); Stoyanova and Lutz, 2021 [DOI](#)), ghrelin influences neurogenesis in the V-SVZ and subgranular zone of the hippocampal dentate gyrus (Hornsby et al., 2016 [DOI](#); Kent et al., 2015 [DOI](#); Kim et al., 2015 [DOI](#); Li et al., 2014 [DOI](#), 2013 [DOI](#); Moon et al., 2009 [DOI](#)). However, its role in blood vessel-guided neuronal migration has not been studied.

To elucidate the effects of blood flow on blood vessel-guided cell migration, we focused on neuronal migration along blood vessels in the adult RMS and OB. The results showed that new neurons migrate along blood vessels with high blood flow and that their migration is affected by changes in blood flow. We also found that ghrelin promotes neuronal migration through activation of actin cytoskeleton contraction at the rear of the cell soma, indicating that blood flow influences neuronal migration via ghrelin signaling. Furthermore, we found that calorie restriction promotes the migration of OB neurons, suggesting that the blood flow-dependent mechanism of neuronal migration could be part of a system to improve sensory function in response to physiological change in the body.

Results

New neurons migrate along blood vessels with abundant blood flow

Previous studies have revealed that V-SVZ-derived new neurons migrate along blood vessels in the RMS and GCL (Bovetti et al., 2007 [DOI](#); Snapyan et al., 2009 [DOI](#); Whitman et al., 2009 [DOI](#)). However, the neuron-vessel interactions along the entire migration route remain unknown.

Therefore, we first studied blood vessel-guided neuronal migration in the RMS and OB using three-dimensional imaging in 6 to 12-week-old adult mice, which enables analysis of the *in vivo* spatial relationship between new neurons and blood vessels. We measured the distance from the soma of new neurons, labeled with green fluorescent protein (GFP)-encoding adenovirus, to the nearest blood vessel labeled with RITC-Dex-GMA. The results showed that GFP⁺ new neurons interact closely with blood vessels in the RMS, GCL, external plexiform layer, and GL (**Figure 1A** [DOI](#)–D, Supplemental Movie S1). In particular, in the RMS and GL, the majority of cells were present within 5 μ m of the inner surface of vessels (**Figure 1D** [DOI](#)), suggesting that new neurons in these regions frequently use blood vessels as migration scaffolds. To visualize blood vessels, we also used *Flt1-DsRed* transgenic mice, in which vascular endothelial cells were specifically labeled with DsRed (Matsumoto et al., 2012 [DOI](#)). Using *Dcx-EGFP/Flt1-DsRed* double transgenic mice, we observed close spatial relationships between new neurons and blood vessels (Supplemental Movies S2 and S3). Transmission electron microscopy revealed direct attachment of migratory neurons, identified as cells with an elongated cell body, a dark cytoplasm with many free ribosomes, and an electron-dense nucleus with multiple nucleoli (Doetsch et al., 1997 [DOI](#)), to thin astrocytic endfeet enwrapping blood vessels (**Figure 1E** [DOI](#)), as previously reported in the RMS and GCL (Bovetti et al., 2007 [DOI](#); Whitman et al., 2009 [DOI](#)). These results indicate that new neurons migrate along blood vessels through their entire migration route, suggesting the possibility that their movement may be influenced by blood flow.

To examine the relationship between neuronal migration and blood flow, we recorded movement of new neurons and red blood cell (RBC) flow using two-photon laser scanning microscopy. Neuronal migration was recorded in the GL of *Dcx-EGFP* mice, in which new neurons were labeled with enhanced GFP (EGFP) (**Figure 1F** [DOI](#)). RBC flow was also recorded in each vessel segment in the GL by line-scan measurements (**Figure 1F** [DOI](#)). We identified migrating new neurons as EGFP⁺ cells whose position changed during the imaging period in *Dcx-EGFP* mice, in which the majority of EGFP⁺ cells are stationary. Most EGFP⁺ migrating cells (90.2%) were located within 10 μ m of the nearest vessel (**Figure 1G** [DOI](#)). To compare the blood vessel-guided migration of new neurons among blood vessels with different flows, we classified the vessels into two groups, high-flow and low-flow vessels, using the median RBC flow velocity as the criterion (**Figure 1H** [DOI](#)). New neurons were more abundant near high-flow vessels than near low-flow vessels (**Figure 1H** [DOI](#)). These cells migrated at a small angle to the longitudinal axis of blood vessels (Figure S1), indicating that new neurons use blood vessels as migration scaffolds. Next, we compared the migration speed of new neurons between the low- and high-flow perivascular region, the area within 10 μ m from the nearest vessel. Migration speed, as measured by the migration distance of the cell soma, was significantly higher in the perivascular region of high-flow vessels than in that of low-flow vessels (**Figure 1I** [DOI](#)). A previous study showed that new neurons undergo discontinuous migration, including migratory and stationary periods in the GL (Liang et al., 2016 [DOI](#)). The proportion of the migratory period of new neurons in the total imaging period was larger in high-flow perivascular regions than in low-flow regions (**Figure 1J** [DOI](#)). Furthermore, the maximum migration speed, calculated from two consecutive imaging frames, was significantly higher for neurons migrating along high-flow vessels than for those migrating along low-flow vessels (**Figure 1K** [DOI](#)). These results suggest that neuronal migration is promoted in perivascular regions with abundant blood flow.

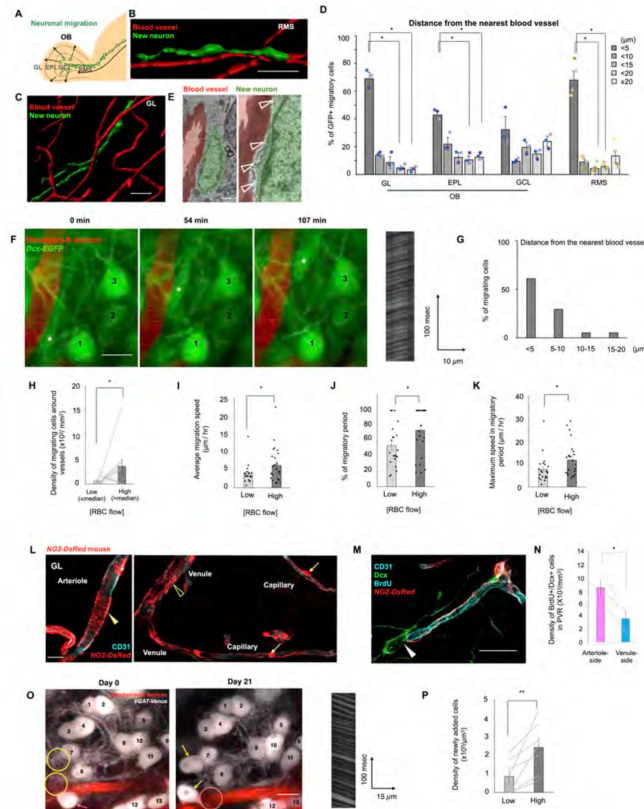


Figure 1

New neurons migrate along blood vessels with abundant flow in the adult brain.

- (A) Experimental scheme.
- (B, C) Three-dimensional reconstructed images of a new neuron (green) and blood vessels (red) in the rostral migratory stream (RMS) (B) and glomerular layer (GL) (C) of 6- to 12-week-old adult mice.
- (D) Distance between new neurons and nearest vessels in the olfactory bulb and RMS (one-way repeated measures ANOVA followed by Bonferroni's test; 3 and 4 mice for the analysis in the OB and RMS, respectively).
- (E) Transmission electron microscopy image of a new neuron (green) in close contact with a blood vessel (red) in the GL of a 6- to 12-week-old adult mouse. Astrocytes (clear arrowheads).
- (F) Time-lapse images of a migrating neuron (indicated by asterisks) in the GL of a 6- to 12-week-old *Dcx-EGFP* mouse. Red blood cell (RBC) flow is recorded as a two-photon line-scan image as shown in the right panel. Stationary cells are indicated by sequential numbers.
- (G) Average distance between migrating cells and nearest blood vessels (41 cells from 38 mice).
- (H) Density of perivascular migrating cells (Wilcoxon signed-rank test; 10 mice).
- (I) Average migration speed (Welch's *t*-test; low, 19 cells, high, 25 cells from 39 mice).
- (J) Percentage of migratory period (Mann-Whitney U-test; low, 19 cells, high, 24 cells from 38 mice).
- (K) Maximum migration speed (unpaired *t*-test; low, 22 cells, high, 24 cells from 39 mice).
- (L) Fluorescent images in the *NG2-DsRed* mouse GL. Arterioles, capillaries, and venules were characterized by band-like smooth muscle cells (solid arrowhead), pericytes (arrows), and fenestrated smooth muscle cells (clear arrowhead), respectively. CD31 (blue), DsRed (red).
- (M) Fluorescent image of a *Dcx*+/*BrdU*+ new neuron (solid arrowhead) attached to a capillary. *Dcx* (green), *BrdU* (blue), CD31 (blue, tube-like structures), DsRed (red).
- (N) Density of *BrdU*+/*Dcx*+ cells in the perivascular region of arteriole-side and venule-side capillaries (paired *t*-test; 3 mice).
- (O) Two-photon images of GABAergic neurons (white) and a blood vessel (red) in the *VGAT-Venus* mouse GL. Circles show positions of added (yellow) and lost (pink) Venus+ cells. Added and lost neurons are indicated by yellow and pink arrows, respectively. RBC flow on Day 21 is shown in the right panel.
- (P) Density of newly added neurons in the perivascular region (paired *t*-test; 7 mice). Data are presented as the means $\pm$ standard error of the mean (SEM). Scale bars: B, 30 μ m; C, 40 μ m; E, 1 μ m; F, 10 μ m; M, 20 μ m; N, 20 μ m; P, 10 μ m. See also Figure S1 and S2.

To compare the distribution of new neurons in perivascular regions with different blood flows in deep brain regions, we used endomucin, which has been reported to be downregulated by shear stress on vascular endothelial cells *in vitro* (Zahr et al., 2016). Endomucin-negative vessels showed higher RBC flow than endomucin-positive vessels (Figure S2A–C), indicating that endomucin can be used as a marker for RBC flow velocity *in vivo*. The density of bromodeoxyuridine (BrdU)-labeled doublecortin (Dcx)+ new neurons was higher in the perivascular region of endomucin-negative vessels than in that of endomucin-positive vessels in the GL, GCL, and RMS (Figure S2D–G), suggesting that new neurons migrate along vessels with faster blood flow throughout their migration route.

We next compared the distribution of new neurons near arteriole-side and venule-side vessels identified using SLC16A1, which is expressed in venules and venule-side capillaries (Vanlandewijck et al., 2018). The density of perivascular Dcx+/BrdU+ new neurons was significantly higher in SLC16A1-negative perivascular areas than in SLC16A1-positive areas (Figure S3A, B), indicating that they actively migrate in the vicinity of arteriole-side vessels (Figure S3C). To further examine the localization of perivascular new neurons, we analyzed the cell distribution in *NG2-DsRed* mice (Zhu et al., 2008), in which we could identify arterioles, venules, and capillaries based on the morphological differences in NG2+ mural cells as performed previously (Hartmann et al., 2015; Hill et al., 2015). Arterioles, capillaries, and venules were characterized by band-like smooth muscle cells, pericytes, and fenestrated-shaped smooth muscle cells, respectively (Figure 1L). A large proportion of Dcx+/BrdU+ new neurons was observed near capillaries in the GL (arterioles; $11.8 \pm 2.39\%$, venules; $4.34 \pm 0.212\%$, capillaries; $83.8 \pm 2.25\%$) (Figure 1M). The frequency of perivascular neuronal migration was compared in arteriole-side and venule-side capillaries (defined as capillaries one to three order branches away from the nearest arterioles and venules, respectively). The density of Dcx+/BrdU+ cells was higher near arteriole-side capillaries than that near venule-side capillaries (Figure 1N). Taken together, these data suggest that most neuronal migration occurs near capillaries and that new neurons prefer capillaries on the arteriole side to those on the venule side (Figure S3C).

Because blood vessel-guided neuronal migration in the adult brain is a conserved phenomenon across species (Akter et al., 2021; Kishimoto et al., 2011; Shvedov et al., 2024), we hypothesized that blood flow may also influence neuronal migration in other brain regions of primates. The neocortex, which supports higher-order brain functions and has undergone evolutionary expansion in primates, was selected as a target region. In common marmosets, but not in mice, V-SVZ-derived new neurons migrate toward the neocortex and ventral striatum (Akter et al., 2021) (Supplemental Movies S4 and S5). In the ventral striatum of 3 to 4-month-old common marmosets, Dcx+ immature neurons localized more frequently near SLC16A1-negative vessels than near SLC16A1-positive vessels, regardless of migration modes (Figure S3D, E). A similar tendency was also observed in the neocortex (Figure S3F). These results suggest that immature neurons prefer to migrate along arteriole-side vessels in the common marmoset brain, and this phenomenon is common between rodents and primates.

Finally, we examined whether new neurons mature near high-flow vessels following their migration along these blood vessels. Neuronal maturation and blood flow were recorded in *VGAT-Venus* mice, in which GABAergic neurons are labeled with Venus. Maturation of new neurons was confirmed by observation of the same regions with an interval of 21 days (Figure 1O). The density of newly added neurons was significantly higher near high-flow vessels than near low-flow vessels (Figure 1P). Consistently, BrdU-labeled mature neurons were frequently observed in the perivascular region of endomucin-negative vessels (Figure S2H). These results suggest that blood vessel-guided neuronal migration supplies new neurons in regions of high blood flow.

Decreases in blood flow affect neuronal migration

To investigate the effects of blood flow on neuronal migration, we performed bilateral carotid artery stenosis (BCAS), which decreases blood flow in the anterior portion of the brain (Hattori et al., 2016; Shibata et al., 2004). Lentiviruses expressing Venus were injected into the V-SVZ to label new neurons migrating toward the OB (Figure 2A). The proportion of Dcx+/Venus+ cells in the OB was significantly lower in the BCAS group compared with the Sham-operated group at 5 dpi (Figure 2B, C), suggesting that blood flow reduction impairs the tangential migration of new neurons in the RMS. To evaluate whether the reduced number of new neurons in the OB following BCAS (Figure 2B, C) is solely due to impaired migration, we examined cell proliferation and survival in the V-SVZ and RMS. Specifically, we quantified the density of Ki67+ proliferating cells and cleaved caspase-3+ apoptotic cells in the sham and BCAS groups. BCAS significantly decreased cell proliferation and increased cell death in both the V-SVZ and RMS (Figure S4), suggesting that reduced neurogenesis and/or survival may contribute to the decreased distribution of new neurons in the OB. To further examine the influence of blood flow changes on neuronal migration, we induced photothrombotic clot formation to reduce blood flow in individual vessels (Figure 2D). Migration of EGFP+ cells along vessels in *Dcx-EGFP* mice was recorded using two-photon imaging (Figure 2E). To prevent effects other than blood flow inhibition, clot formation was induced in upstream vessel fragments distant from vessels close to migrating neurons (Figure 2D). For clot formation, a restricted area of selected vessels was irradiated by a two-photon laser immediately after intravenous injection of rose bengal. Clot formation resulted in blood flow termination in downstream vessels (Figure 2F), which was followed by a decrease in the migration speed of EGFP+ new neurons along downstream vessels (Figure 2E, H). Laser irradiation without rose bengal did not affect the speed of neuronal migration (Figure 2G), indicating that the inhibition of neuronal migration was not due to laser irradiation. These data suggest that blood flow facilitates neuronal migration in the RMS and OB and that blood contains factors influencing neuronal migration.

Ghrelin increases neuronal migration speed by promoting somal translocation

We focused on ghrelin, which can be delivered from the bloodstream to the brain parenchyma, including the OB tissue, by transcytosis across vascular walls (Rhea et al., 2018). In addition, a previous study showed that newly generated neurons expressed growth hormone secretagogue receptor 1a (GHSR1a), a ghrelin receptor, in the V-SVZ, RMS, and OB (Li et al., 2014). Consistent with this report, we detected *Ghsr1a* mRNA in Dcx+ new neurons in the VSVZ, RMS, and OB (Figure S5A). A previous study showed that migration of V-SVZ-derived new neurons is attenuated in ghrelin knockout mice (Li et al., 2014), suggesting that ghrelin stimulates neuronal migration. At first, to examine transcytosis of ghrelin in the OB, we introduced fluorescently labeled ghrelin into the bloodstream. We found accumulation of fluorescent ghrelin in the RMS and OB as reported previously (Rhea et al., 2018) (Figure 3A, B, Figure S6).

Fluorescence signals were observed in vascular endothelial cells and parenchymal tissue in the RMS and OB (Figure 3B), indicating that blood-derived ghrelin crosses the vascular wall into the brain parenchyma and is delivered to new neurons. In addition, we observed high levels of fluorescent signal in vascular endothelial cells of endomucin-negative, high-flow vessels (Figure 3C, D), which suggests that transcytosis of blood-derived ghrelin may occur more frequently in these vessels, potentially due to increased endothelial endocytosis. We found that some, but not all, vessels showed particularly strong fluorescent signals in parenchymal regions adjacent to the abluminal side of vascular endothelial cells, as visualized by CD31 immunostaining (Feng et al., 2004) (Figure 3A, A"). To quantify this observation, we defined two regions of interest: Area I (perivascular area), within 10 µm of the abluminal surface of CD31-positive endothelium; and Area II (distant area), located 10–20 µm away (Figure 3E). Of note, Area I corresponds to the

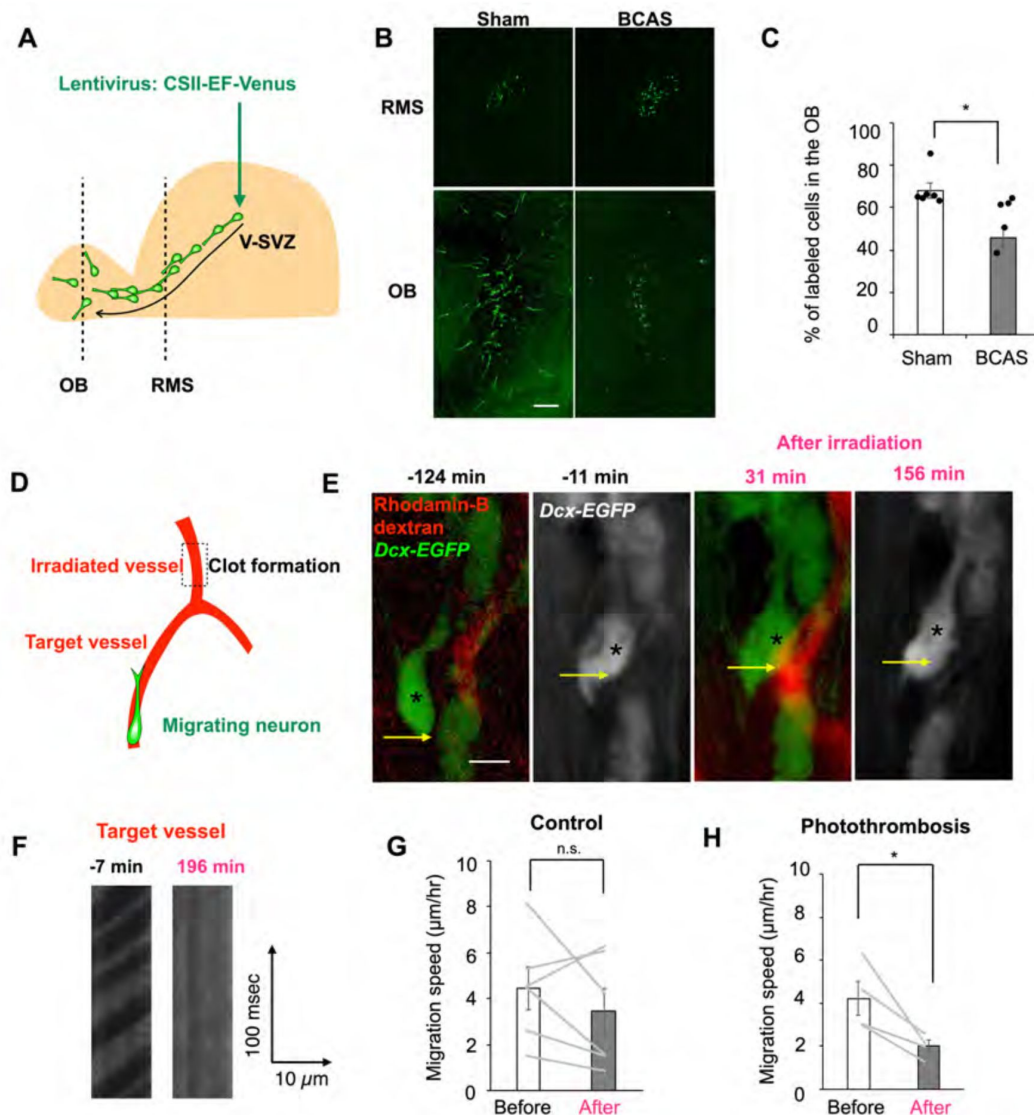


Figure 2

Blood flow inhibition attenuates neuronal migration.

(A, D) Experimental schemes.

(B) Fluorescent images of Venus+ new neurons (green) in the rostral migratory stream and olfactory bulb (OB).

(C) Proportion of Venus+ cells in the OB in the Sham and bilateral carotid artery stenosis groups (unpaired *t*-test; Sham, 6 mice, BCAS, 5 mice).

(E) Two-photon images of neuronal migration (arrows) before and after photothrombotic clot formation in a *Dcx-EGFP* mouse. A new neuron (green), a blood vessel (red).

(F) Line-scan images from a blood vessel shown in (E).

(G, H) Comparison of migration speed before and after laser irradiation in the control (G) (paired *t*-test; 7 cells from 7 mice) and photothrombosis groups (H) (paired *t*-test; 4 cells from 4 mice). Data are presented as the means $\pm$ SEM. Scale bars: B, 100 μ m; E, 10 μ m.

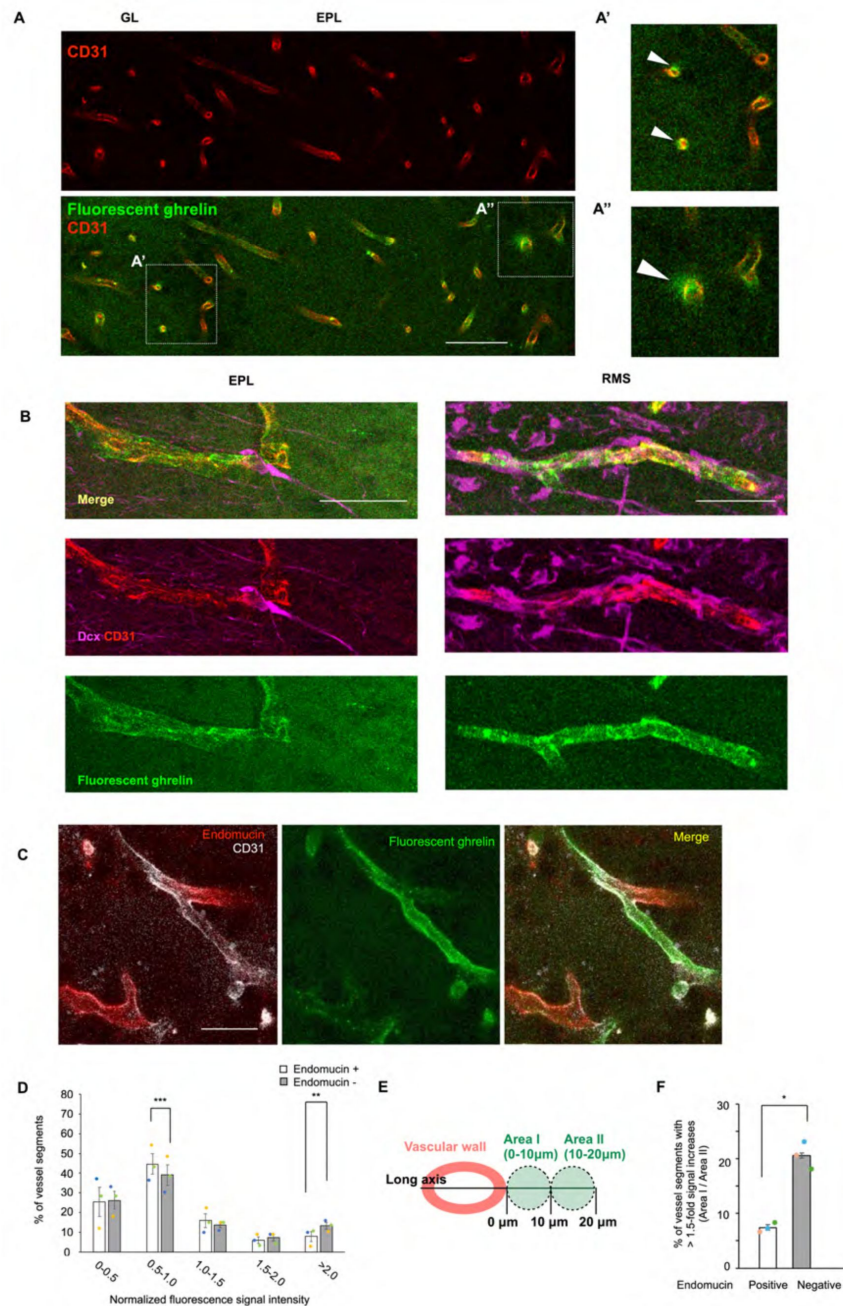


Figure 3

Ghrelin is delivered from the bloodstream to the RMS and OB in the adult brain.

(A) Representative images of the OB and the cortex of a fluorescent ghrelin-infused mouse (6-to 12-week-old). CD31 (red), Dcx (magenta), fluorescent ghrelin (green). (A', A'') High magnification images of boxed areas in (A). Arrowheads indicate fluorescent signals in parenchymal areas adjacent to the vascular endothelium.

(B) Fluorescent images of neuronal migration along blood vessels in the EPL and the RMS. CD31 (red), Dcx (magenta), fluorescent ghrelin (green).

(C) Fluorescent images of blood vessels in the GL (C). CD31 (white), endomucin (red), fluorescent ghrelin (green).

(D) Normalized fluorescence signal intensity in vascular endothelial cells (paired *t*-test; 3 mice).

(E) Schematic diagram for analyzing signal gradients in extra-vascular areas.

(F) Percentage of vessel segments with >1.5-fold increases in Area I relative to Area II (paired *t*-test; 3 mice). Data are presented as the means $\pm$ SEM. Scale bars: A, 50 μ m; B, 20 μ m (EPL), 10 μ m (RMS); C, 20 μ m. See also Figure. S3.

perivascular region where new neurons are frequently observed (**Figure 1**). To determine whether ghrelin transcytosis occurs more frequently in high-flow vessels, we quantified signal gradients in the extra-vessel regions as fold changes (Area I / Area II), as illustrated in **Figure 3E**. The proportion of vessel segments with >1.5-fold increases was significantly higher in endomucin-negative vessels than in endomucin-positive ones (**Figure 3F**). Furthermore, vessel segments with >2-fold increases were observed exclusively in the endomucin-negative group ($6.48\% \pm 1.18\%$). These data suggest that, in high-flow vessels, blood-derived ghrelin accumulates more in the immediate perivascular region than more distant areas. This supports the possibility that elevated blood flow increases the delivery of ghrelin to the vascular endothelium, enhancing its transcytosis into adjacent brain parenchyma. This mechanism may underlie the preferential migration of new neurons along high-flow perivascular regions, as shown in **Figure 1**.

To assess the direct effect of ghrelin on neuronal migration, we applied recombinant ghrelin to V-SVZ cultures, in which new neurons emerge and migrate as chains (**Figure 4A**). Ghrelin significantly increased the migration distance of these neurons (**Figure 4B**), indicating enhanced chain migration. We then used super-resolution time-lapse imaging to examine individually migrating neurons with or without knockdown (KD) of growth hormone secretagogue receptor 1a (GHSR1a), a ghrelin receptor expressed in V-SVZ-derived new neurons (Li et al., 2014) (**Figure 4C**). Ghrelin enhanced the migration speed of control cells (*lacZ*-KD) cells, indicating that it also facilitates individual migration (**Figure 4D**). V-SVZ-derived new neurons exhibit saltatory migration consisting of a migratory phase and a resting phase (Ota et al., 2014).

Ghrelin application increased the migratory phase proportion (**Figure 4E**) but not the length of the migration cycle (**Figure 4F**), suggesting that ghrelin signaling elongates the migratory phase of neuronal migration. Cultured new neurons alternate between leading process extension and somal translocation (**Figure 4C**). Ghrelin application did not affect the length or speed of leading process extensions (**Figure 4G**, H). In contrast, the somal translocation speed and somal stride length were significantly increased by ghrelin application (**Figure 4I**, J). No such effects were observed in *Ghsr1a*-KD cells (**Figure 4D**, E, I, J), suggesting that ghrelin promotes neuronal migration through *ghsr1a*. Taken together, these results suggest that ghrelin signaling promotes somal translocation and thus increases the efficacy of neuronal migration.

Ghrelin signaling has been reported to regulate actin cytoskeletal dynamics in astrocytoma cells (Dixit et al., 2006), which led us to examine whether a similar mechanism operates in migrating neurons. Somal translocation of migrating cortical interneurons is driven by formation of the actin cup, an accumulation of F-actin at the rear of the cell soma (Martini and Valdeolmillos, 2010). Time-lapse imaging of cultured new neurons expressing EGFP fused to the calponin homology domain of utrophin (EGFP-UtrCH), a fluorescent reporter for F-actin (Burkel et al., 2007), revealed discontinuous formation of the actin cup at the rear of the cell soma (**Figure 4K**). Ghrelin application extended the duration of actin cup formation (**Figure 4L**) and increased the migration distance during actin cup formation (**Figure 4M**, Supplemental Movie S6). These effects were not observed in *Ghsr1a*-KD cells (**Figure 4L**, M, Supplemental Movie S7), suggesting that GHSR1a-mediated ghrelin signaling promotes somal translocation of new neurons by activation of actin cup formation at the rear of the cell soma.

To investigate the effects of ghrelin signaling on neuronal migration in the OB *in vivo*, a mixture of lentiviruses encoding either *Ghsr1a*-KD or control shRNA was injected into the OB core of the same mouse (**Figure 5A**). Of the total labeled Dcx⁺ cells, the percentage of Dcx⁺ cells reaching the GL was significantly lower in the *Ghsr1a*-KD group than that in the control group (**Figure 5B**, C), suggesting that ghrelin enhances individual radial migration of new neurons in the OB. Coinjection of control and *Ghsr1a*-KD lentiviruses into the same site allowed us to directly compare their effects under identical conditions. However, this approach may allow reciprocal interactions between neurons infected with different constructs, potentially confounding cell-autonomous effects. To address this, we also performed separate injections of control and *Ghsr1a*-

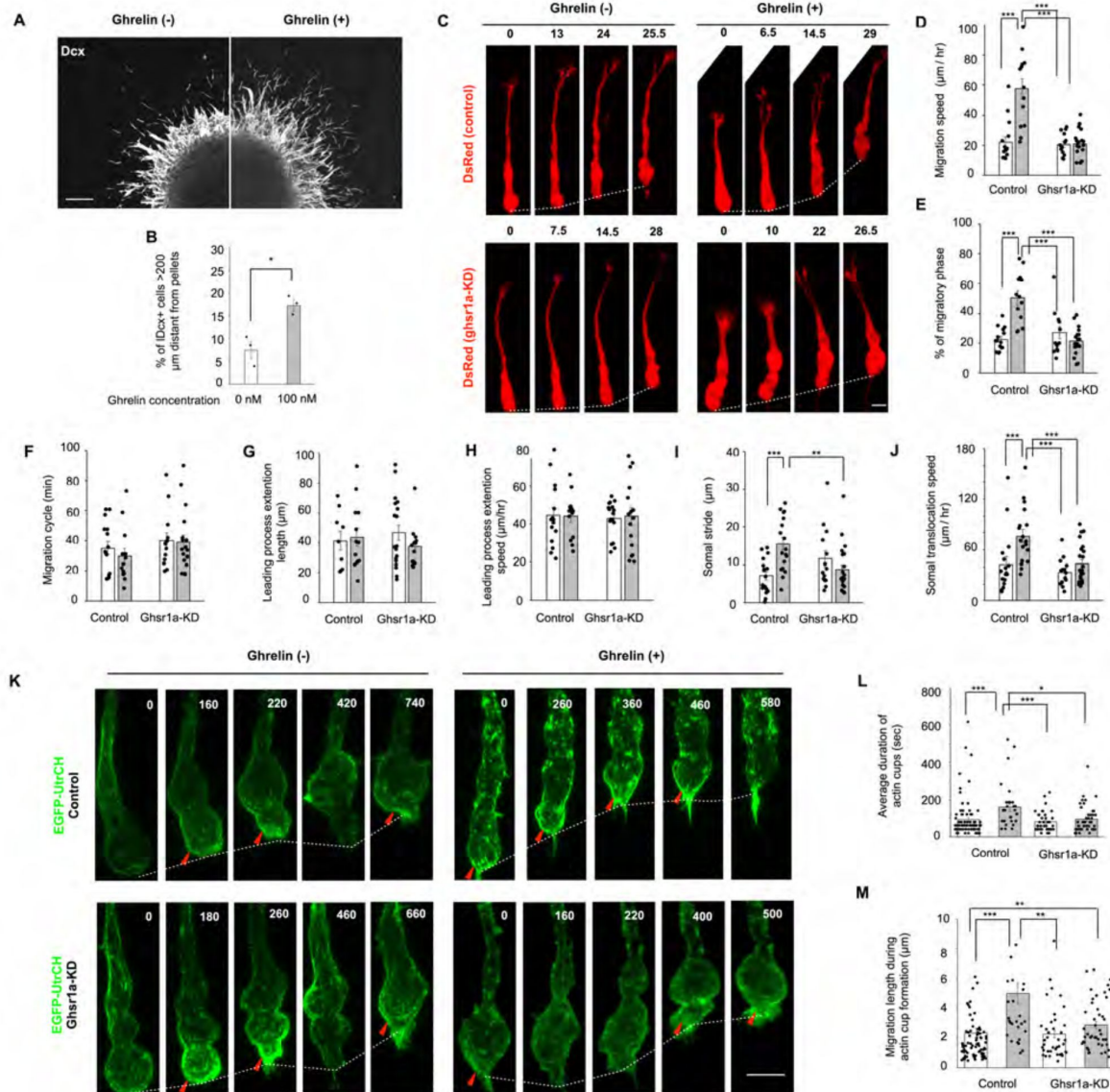


Figure 4

Ghrelin promotes neuronal migration by activation of actin cup formation.

(A) Fluorescent images of Matrigel culture. Dcx (white).

(B) Percentage of Dcx+ cells > 200 μm distant from the edge of pellets (unpaired *t*-test; 3 independent cultures prepared on different days).

(C) Time-lapse images of cultured new neurons expressing DsRed (red). The number above each panel indicates minutes after initiation of migration.

(D–J) Migration speed (D), percentage of migratory phase (E), migration cycle (F), length/speed of leading process extension (G, H), and stride/speed of somal translocation (I, J) in neuronal migration (one-way ANOVA followed by Turkey-Kramer test; D–F: control / Ghrelin (-), 15 cells, control / Ghrelin (+), 13 cells, KD / Ghrelin (-), 13 cells, KD / Ghrelin (+), 18 cells, I, J: control / Ghrelin (-), 17 events, control / Ghrelin (+), 18 events, KD / Ghrelin (-), 14 events, KD / Ghrelin (+), 23 events).

(K) Time-lapse images of actin cup formation (arrowheads) in the cell soma of new neurons. EGFP-UtrCH (green). Condensed dots of F-actin were scattered throughout the elongated cell soma in a control cell with ghrelin application.

(L, M) Average duration of actin cups (N) and migration distance during actin cup formation (O) in new neurons (Kruskal-Wallis test followed by the Steel-Dwass test; control / Ghrelin (-), 79 cells, control / Ghrelin (+), 31 cells, KD / Ghrelin (-), 39 cells, KD / Ghrelin (+), 44 cells). Data are presented as the means $\pm$ SEM. Scale bars: A, 100 μm ; C, 5 μm ; K, 5 μm .

KD lentiviruses into different mice (Figure S7A). Consistent with the co-injection results, *Ghsr1a*-KD cells showed reduced distribution in the GL compared with that in control cells (Figure S7B), which supports that ghrelin signaling enhances neuronal migration in a cell-autonomous manner. When lentiviruses were injected into the V-SVZ (Figure 5D), *Ghsr1a*-KD (DsRed+) cells exhibited decreased distribution in the GL (Figure 5E, F) and increased distribution in the RMS compared with control (EmGFP+) cells (Figure 5G, H). These data indicate that ghrelin signaling facilitates both individual migration in the OB and chain migration in the RMS.

To determine whether the altered distribution of new neurons after *Ghsr1a*-KD is due to impaired migration rather than changes in cell production or survival, we assessed the effects of *Ghsr1a*-KD on the proliferation and survival of new neurons and their progenitors, which express GHSR1a (Li et al., 2014). We quantified the proportion of cleaved caspase-3+ cells and Ki67+ cells from the total labeled cells in the V-SVZ and RMS in both control and *Ghsr1a*-KD groups. We found no significant difference in cleaved caspase-3+ cell proportions between the groups (Control: 874 cells from 5 mice; *Ghsr1a*-KD: 678 cells from 7 mice), suggesting that ghrelin signaling does not influence the survival of new neurons and their progenitors. Similarly, the percentage of Ki67+ cells in the RMS was similar between the two groups (Figure S8), indicating that *Ghsr1a*-KD does not impair cell proliferation in the RMS. However, technical limitations prevented a reliable evaluation of proliferation in the V-SVZ, as lentivirus injection into this region may interfere with GHSR1a expression in not only neural progenitors and new neurons, but also other *ghsr1a*-expressing cell types (Zigman et al., 2006). Although ghrelin signaling has been reported to promote cell proliferation in the V-SVZ (Li et al., 2014), our complementary in vitro KD experiments (Figure 4C–J) and in vivo OB-core lentivirus injections (Figure 5A–C), which did not affect the V-SVZ, consistently demonstrated that *Ghsr1a*-KD reduces neuronal migration. Taken together, our results suggest that blood-derived ghrelin enhances neuronal migration in the RMS and OB by stimulating actin cytoskeleton contraction in the cell soma, rather than by altering cell proliferation or survival.

As shown in Figure 3, ghrelin transcytosis preferentially occurs in high-flow blood vessels, indicating that ghrelin contributes to blood flow-dependent neuronal migration. However, the extent to which this process depends on ghrelin signaling remained unclear. To investigate this, we combined *Ghsr1a*-KD with BCAS. We found that BCAS reduced the percentage of *Ghsr1a*-KD new neurons reaching the OB, similar to the effect seen in control neurons (Figure S9B, see also Figure 2A–C). This suggests that blood flow influences neuronal migration even under *Ghsr1a*-KD conditions. Furthermore, we analyzed the distribution of *Ghsr1a*-KD neurons with respect to vessel flow characteristics. Even under *Ghsr1a*-KD, a higher proportion of new neurons were located in the area of endomucin-negative (high-flow) vessels compared with endomucin-positive (low-flow) vessels (Figure S9C), indicating that *Ghsr1a*-KD does not abolish the preferential association of migrating neurons with high flow vessels. These findings suggest that ghrelin signaling is involved, but is not essential, for the blood flow-dependent guidance of migrating neurons, and that additional blood-derived signals may contribute to this process.

Finally, we examined whether calorie restriction, which has been reported to increase blood ghrelin levels (Toshinai et al., 2001; Tschöp et al., 2000), affects neuronal migration. We used a 70% calorie restriction (CR) protocol, which was previously reported to enhance hippocampal neurogenesis over 14 days (Hornsby et al., 2016). In our study, mice were fed 70% of their daily ad libitum (AL) food intake levels for 5 days (Figure 5I, Figure S10A). The proportion of labeled cells in the GL (Figure 5J) and NeuN+/Dcx-cells among BrdU+ cells in the OB (Figure S10B, C) was larger in the calorie restriction group than in the ad libitum group. However, there was no significant difference in the proportion of *Ghsr1a*-KD cells in the GL between the control and calorie restriction groups (Figure 5K). We found no significant difference in the *Ghsr1a* expression level in OB Dcx+ cells between the AL and CR groups (Figure S5B). Taken together,

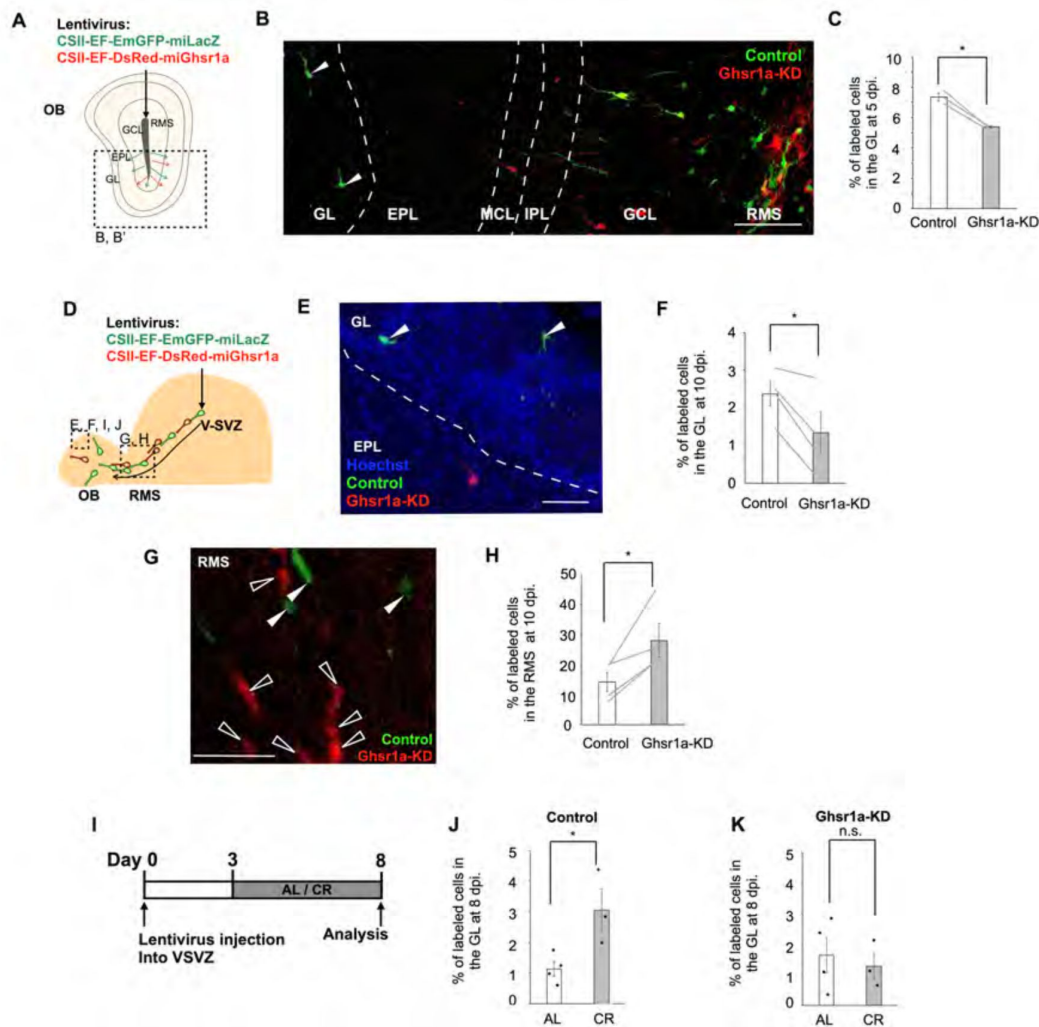


Figure 5

Ghrelin signaling promotes neuronal migration in the adult brain.

(A, D) Experimental schemes. Lentivirus injection into the OB core (A) and the VSVZ (D) was performed in 6- to 12-week-old adult mice.

(B) Fluorescent images of new neurons in the olfactory bulb in (A).

(C) Proportion of labeled cells in the GL at 5 dpi. in (A) (paired *t*-test; 3 mice).

(E, G) Fluorescent images of new neurons in the glomerular layer (GL) (E) and rostral migratory stream (RMS) (G) for the experiments shown in (D). Control cells (white arrowheads), *Ghrelin1a*-KD cells (clear arrowheads).

(F, H) Proportion of labeled cells in the GL (F) and the RMS (H) at 10 dpi. in (D) (paired *t*-test; 4 mice).

(I) Experimental scheme.

(J, K) Proportion of labeled cells in the GL at 8 dpi in the ad libitum (I) and calorie restriction (J) groups (Control, unpaired *t*-test; AL, 4 mice, DR, 3 mice) (KD, unpaired *t*-test; AL, 4 mice, CR, 3 mice). Control (green), *Ghrelin1a*-KD (red). GL (glomerular layer), EPL (external plexiform layer), MCL (mitral cell layer), IPL (internal plexiform layer), GCL (granule cell layer), RMS (rostral migratory stream). AL (ad libitum), CR (calorie restriction). Data are presented as the means ± SEM. Scale bars: B, 100 μm; E, 40 μm; G, 40 μm.

these data suggest that blood flow promotes the migration of OB neurons during starvation via ghrelin signaling and that promoted migration of new neurons increases the number of mature neurons in the OB.

Discussion

Previous studies have shown the role of blood vessels as physical scaffolds in the migratory routes of new neurons in various situations (Bovetti et al., 2007 [DOI](#); Fujioka et al., 2017 [DOI](#); Grade et al., 2013 [DOI](#); Kojima et al., 2010 [DOI](#); Ohab et al., 2006 [DOI](#); Snapyan et al., 2009 [DOI](#); Sun et al., 2015 [DOI](#); Thored et al., 2007 [DOI](#); Whitman et al., 2009 [DOI](#); Yamashita et al., 2006 [DOI](#)). Proteins expressed by vascular cells have been reported to facilitate neuronal migration by binding to transmembrane receptors of new neurons (Fujioka et al., 2017 [DOI](#); Grade et al., 2013 [DOI](#); Ohab et al., 2006 [DOI](#); Snapyan et al., 2009 [DOI](#)). However, whether neuronal migration is affected by blood flow remains unknown. Therefore, in this study, we focused on the specific feature of the vasculature as a pipeline for blood delivery. The effects of blood flow on neuronal migration are difficult to detect with previously used experimental procedures such as immunohistochemistry of vascular endothelial cell markers in fixed tissue sections and time-lapse live imaging of brain slice cultures. Therefore, we classified blood vessels using molecular markers that reflect blood flow properties and performed *in vivo* live imaging to record blood flow. This approach enabled us to reveal the effects of blood flow on neuronal migration.

Previous studies reported that new neurons migrate along blood vessels in the RMS and GCL of the OB (Bovetti et al., 2007 [DOI](#); Snapyan et al., 2009 [DOI](#); Whitman et al., 2009 [DOI](#)). In the present study, three-dimensional imaging was performed over a wide area across the entire migration route in transparent brains, where the positional relationship between new neurons and blood vessels can be observed. The results suggest that new neurons use blood vessels as migration scaffolds throughout their migration route (**Figure 1A–D** [DOI](#)). The distance we found between new neurons and blood vessels was larger than that reported in a previous study (Snapyan et al., 2009 [DOI](#)). This might be due to our method of measuring the distance from the blood vessels to the cell soma of new neurons, rather than the distance to the entire new neurons. To investigate the effects of blood flow on blood vessel-guided neuronal migration, we used two-photon imaging to record neuronal migration and blood flow *in vivo* in the GL in the superficial area of the OB, which is a useful model for analyzing blood vessel-guided neuronal migration. The results suggest that new neurons migrate faster near blood vessels with high flow than in the areas of those with low flow. Neural stem cells have been reported to increase blood flow in the adult V-SVZ (Lacar et al., 2012 [DOI](#)), raising the possibility that new neurons may increase blood flow in the OB. However, our observation that inhibition of blood flow suppressed neuronal migration suggests that blood flow facilitates neuronal migration. New neurons terminated their migration and differentiated into mature interneurons in perivascular regions with abundant flow.

Thus, the blood flow-dependent mechanism of neuronal migration may supply new neurons to areas appropriate for their function.

We found that the migration speed of new neurons was decreased by blood flow reduction, suggesting that neuronal migration depends on blood flow in the adult brain. Although we cannot exclude the possibility that BCAS alters the cell proliferation or survival of new neurons, our photothrombotic clot formation experiments are better suited to directly examine how acute reductions in blood flow influence neuronal migration. These experiments allowed us to assess the migration speed of new neurons shortly after inducing localized blood flow inhibition. Based on the finding that neuronal migration was attenuated in our BCAS and photothrombosis experiments, we hypothesized that a blood-derived factor facilitates neuronal migration. A previous study demonstrated that the migration of V-SVZ-derived new neurons was attenuated in ghrelin knockout mice (Li et al., 2014 [DOI](#)). In our study, we found that the migration of cultured new

neurons was enhanced by the application of ghrelin to the culture medium, and this effect was abolished by *Ghsr1a* knockdown (KD). These findings suggest that ghrelin directly stimulates neuronal migration through its receptor, GHSR1a, on new neurons. A previous study showed that GHSR1a is expressed in various regions of the brain (Zigman et al., 2006 [DOI](#)). In our experiments, new neuron-specific KD of *Ghsr1a* indicated that ghrelin signaling acts in a cell-autonomous manner to regulate neuronal migration. Nevertheless, a recent study (Stark et al., 2024 [DOI](#)) showed that GHSR1a was expressed in various cell types, including glutamatergic and GABAergic neurons, suggesting that ghrelin may also exert non-cell-autonomous effects on neuronal migration. Given the presence of diverse cell types—such as neurons, microglia, pericytes, and astrocytes—along the migratory route, it remains possible that GHSR1a activation in these neighboring cells contributes to the overall regulation of neuronal migration.

Furthermore, we identified the cellular and cytoskeletal mechanisms underlying this effect on migration. The results indicate that ghrelin enhances somal translocation during migration by activating actin cytoskeletal dynamics at the rear of the neuronal soma. We also found that ghrelin signaling increases the migration distance of cell soma, which increases the migratory phase duration. Further studies are needed to elucidate how ghrelin promotes actin cup formation in migrating neurons. Given that Rac, a Rho family GTPase, mediates actin remodeling downstream of ghrelin in astrocytoma cells (Dixit et al., 2006 [DOI](#)), it is possible that Rac may also be involved in ghrelin-induced cytoskeletal regulation in new neurons. In addition to actin remodeling, ghrelin may regulate microtubule dynamics. Ghrelin signaling was shown to modulate microtubules via SFK-dependent phosphorylation of α -tubulin (Slomiany and Slomiany, 2017 [DOI](#)), raising the possibility that ghrelin promotes somal translocation of new neurons through coordinated regulation of both actin and microtubule networks (Kaneko et al., 2017 [DOI](#)). These results suggest that ghrelin signaling promotes neuronal migration in the RMS and OB *in vivo*, which could further strengthen our finding that blood flow plays a role in neuronal migration.

Furthermore, we found that blood-derived ghrelin crosses the vascular wall into the RMS and OB. Although we could not exclude the possibility that ghrelin is produced in the brain parenchyma (Howick et al., 2017 [DOI](#)), these results suggest that blood-derived ghrelin is provided to new neurons and promotes somal translocation by activating actin cup formation. Our results indicate that higher blood flow delivers a larger amount of ghrelin to the vascular endothelium resulting in increased ghrelin transcytosis across vascular walls. It is possible that high blood flow increases the amount of ghrelin reaching the luminal surface of vascular endothelial cells, thereby increasing the possibility of ghrelin transcytosis into the brain parenchyma. Our findings suggest that blood-derived ghrelin contributes to the blood flow-dependent neuronal migration.

Because blood ghrelin levels increase during fasting (Toshinai et al., 2001 [DOI](#); Tschöp et al., 2000 [DOI](#)), we examined the possibility that neurogenesis in the OB is affected by feeding conditions. We found that calorie restriction promoted the migration of OB neurons, an effect which was cancelled by *Ghsr1a*-KD, suggesting that calorie restriction facilitates the OB neurogenesis through ghrelin signaling. A recent study demonstrated that fasting elevated *Ghsr1a* expression in the OB (Stark et al., 2024 [DOI](#)), raising the possibility that calorie restriction may have a similar effect. However, in our analysis, the number of *Ghsr1* mRNA puncta in Dcx⁺ new neurons did not differ between the AL and CR groups (Figure S5B), suggesting that calorie restriction does not alter GHSR1a expression levels in new neurons. Although we cannot exclude the possibility that calorie restriction modulates GHSR1a expression in other OB cell types, our combined calorie restriction and *Ghsr1a*-KD experiments support a cell-autonomous contribution of ghrelin signaling to calorie restriction-induced enhancement of neuronal migration. Although our data indicate that ghrelin signaling is essential for fasting-induced acceleration of neuronal migration, calorie restriction may also alter the concentrations of other circulating factors (Alogaiel et al., 2025 [DOI](#); Bonnet et al., 2020 [DOI](#); Wu et al., 2024 [DOI](#)), which could independently influence the behavior of migrating neurons. Since the supply of new neurons to the OB has been suggested to improve olfactory

function in food-seeking behavior (Lazarini and Lledo, 2011 [↗](#)), increased neurogenesis caused by long-term calorie restriction may contribute to improved olfactory function for food seeking during starvation.

The increased speed of somal translocation and elongated duration of the migratory phase in cultured new neurons are in accord with the *in vivo* increase in migration speed and duration of the migratory period of new neurons around high-flow vessels, respectively. Effective somal translocation has been suggested to be advantageous for cell migration in densely packed tissues (Kengaku, 2018 [↗](#)). It is possible that the promotion of somal translocation by ghrelin signaling and blood flow overcomes difficulties in smooth migration of new neurons in dense tissues of the adult brain.

Our data showed that new neurons preferentially migrate along arteriole-side vessels rather than venule-side vessels in both mouse and common marmoset brains, suggesting that the mechanism of blood flow-dependent neuronal migration is conserved across rodent and primate species, as well as across brain regions. A previous study identified a ghrelin homolog in the common marmoset with high sequence similarity to the murine version (Takemi et al., 2016 [↗](#)). In addition, the marmoset GHSR1a homolog shares 95.36 % amino acid identity with that of the mouse (<https://www.ncbi.nlm.nih.gov/protein/380748978> [↗](#)). These findings suggest that blood-derived ghrelin promotes neuronal migration in the common marmoset brain in a manner similar to that in mice. Previous reports have shown that new neurons migrate along blood vessels to damaged areas after brain injury (Fujioka et al., 2017 [↗](#); Grade et al., 2013 [↗](#); Kojima et al., 2010 [↗](#); Ohab et al., 2006 [↗](#); Thored et al., 2007 [↗](#); Yamashita et al., 2006 [↗](#)). Neuronal migration may be influenced by blood flow under pathological conditions as well as during blood vessel-guided migration under the physiological conditions shown in this study. It is possible that blood contains factors in addition to ghrelin that regulate neuronal migration. Blood flow may coordinate biological events between different organs by sending beneficial factors produced outside the brain to influence regionally restricted neuronal migration. Further studies could identify unknown factors involved in the mechanism of blood flow-dependent cell migration, which could contribute to the development of blood flow-based therapies for neurological diseases.

Methods

Animals

All *in vivo* experiments were performed on 6 to 12-week-old C57BL/6J male mice. Wild-type mice were purchased from Japan SLC (Shizuoka, Japan). The following transgenic mice were used:

Dcx-EGFP mice (Gong et al., 2003 [↗](#)) provided by the Mutant Mouse Research Resource Center (MMRRC), *VGAT-Venus* line #39 (Wang et al., 2009 [↗](#)), *Flt1-DsRed* (Matsumoto et al., 2012 [↗](#)) and *NG2-DsRed* (Zhu et al., 2008 [↗](#)). Cells were dissociated from postnatal day 0–1 (P0–P1) pups for *in vitro* experiments. Three to four-month-old postnatal common marmosets were obtained from three mating pairs in a domestic animal colony and used for immunohistochemistry. All animals were housed under a 12-hour light /dark cycle with ad libitum access to food and water. All experiments involving live animals were performed in accordance with the guidelines and regulations of Nagoya City University and the National Institute for Physiological Sciences.

BrdU administration

Bromodeoxyuridine (BrdU, MilliporeSigma), dissolved in sterile phosphate-buffered saline (PBS), was intraperitoneally administered to mice (50 mg/kg) twice with an interval of 2 hours. Mice were fixed at 10 days post injection (dpi) to observe immature neurons or 28 dpi to observe mature olfactory neurons.

Immunohistochemistry

Immunohistochemistry was performed as previously described for brain tissues of mice (Sawada et al., 2011) and common marmosets (Akter et al., 2021). Animals were transcardially perfused with PBS (pH 7.4) followed by 4% paraformaldehyde (PFA) in 0.1 M phosphate buffer (PB). The brains were removed from the skull and postfixed in the same fixative (24 hours for mice, 48 hours for common marmosets). Coronal sections were prepared using a vibratome (VT-1200S; Leica) (50 μ m thick in mice, 60 μ m thick in common marmosets). The sections were incubated with 10% normal donkey serum/0.2% Triton X-100 in PBS (blocking solution) for 30 min at room temperature (RT), the primary antibodies in blocking solution for 24 hours at 4°C, and AlexaFluor-/biotin-conjugated secondary antibodies (1:1000, Invitrogen) in the same solution for 2 hours at RT. For signal amplification, the sections were pretreated with 1% H₂O₂ in PBS for 40 min at RT before blocking. The signals were amplified with biotin-conjugated antibodies and a Vectastain Elite ABC Kit (Vector Laboratories) and visualized using Tyramide Signal Amplification (Thermo Fisher Scientific). For BrdU staining, sections were treated with 1 M HCl at 37°C for 30 min after 1% H₂O₂ treatment. After staining, the sections were mounted with aqueous mounting medium (PermaFluor, Lab Vision Corporation). Z-stack images were obtained using an LSM700 confocal laser scanning microscope (Carl Zeiss) with a 20 $\times$ objective (512 $\times$ 512 pixels, 1.25 μ m per pixel, 1 μ m z-step size). For cell density analysis, the perivascular region was defined as the area within 10 μ m of the edge of CD31⁺ vessels. BrdU administration was performed to identify relatively immature cells in the Dcx⁺ cell population including newly generated cells with different differentiation states.

The following primary antibodies were used: rat anti-GFP (1:500, 04404-84, Nacalai Tesque, Inc.); rabbit anti-GFP (1:500, No. 598, Medical and Biological Laboratories Co., LTD.); rabbit anti-DsRed (1:2000, 632496, Clontech); guinea pig anti-doublecortin (Dcx) (1:400, ab2253, MilliporeSigma); rabbit anti-Dcx (1:500, #4604, Cell Signaling Technology); rat anti-CD31 (1:100, 550274, BD Biosciences); mouse anti-human CD31 (1:100, Dako); rat anti-endomucin (1:500, sc-65495, Santa Cruz Biotechnology); rat anti-BrdU (1:100, ab6326, Abcam); rabbit anti-NeuN (1:1000, ab177487, abcam); rabbit anti-SLC16A1 (1:500, TA321556, Origene); rat anti-Ki67 (1:500, #14-5698-82, eBioscience); and rabbit anti-cleaved caspase-3 (1:200, #9661, Cell Signaling Technology). Nuclei were stained with Hoechst (1:5000, H1399, Thermo Fisher Scientific).

Three-dimensional imaging

For obtaining three-dimensional images from the rostral migratory stream (RMS), new neurons were visualized in *Dcx-EGFP* or *Dcx-EGFP/Flt1-DsRed* mice. Because the population of EGFP⁺ cells includes not only V-SVZ-derived migrating new neurons but also other types of cells in *Dcx-EGFP*, new neurons were labeled with adenoviruses encoding enhanced GFP (Ad-GFP, Vector Biolabs) for the analysis of new neuron-blood vessel interactions. Adenoviruses were stereotactically injected into the V-SVZ (1.0 mm anterior, 1.1 mm lateral to bregma, and 1.6–2.0 mm deep) to label new neurons in the RMS and injected into the RMS (2.8 mm anterior, 0.82 mm lateral to bregma, and 2.8–3.0 mm deep) to label new neurons in the OB. The blood vessel lumen was visualized with RITC-Dex-GMA as previously reported with modifications (Miyawaki et al., 2020). At 5 dpi, mice were transcardially perfused with PBS and 4% PFA/0.1M PB followed by RITC-Dex-GMA. Mouse bodies were incubated in a 37°C water bath for 3 hours for polymerization. The brains were postfixed with SHIELD solutions (Lifecanvas Technologies) as previously reported (Park et al., 2018). Then, they were cleared using SmartClear II Pro (Lifecanvas Technologies). For visualization of new neurons in the common marmoset brain, the brains were incubated in 10% normal donkey serum/ 0.5% Triton X-100 in PBS for 30 min, the anti-Dcx primary antibody in blocking solution for 5 days, and the AlexaFluor-conjugated secondary antibodies (1:1000, Invitrogen) in the same solution for 3 days at 37°C. Refractive index matching was performed using EasyIndex (Lifecanvas Technologies) before imaging. Z-stack images were acquired with a light-sheet fluorescent microscope (Carl Zeiss) with a 5 $\times$ and 20 $\times$ objective (1216 $\times$ 1216 pixels, 1.3

μm per pixel, 1.2–1.4 μm z-step size). Three-dimensional reconstruction was performed using Imaris software (Carl Zeiss). The three-dimensional distance was measured using ZEN software (Carl Zeiss) in light-sheet Z-stack images containing all of the GFP-positive cells in the OB hemisphere per mouse.

Bilateral carotid artery stenosis

Bilateral carotid artery stenosis was performed as previously described with modifications (Hattori et al., 2016 [DOI](#); Shibata et al., 2004 [DOI](#)). After midline incision of the mouse cervical region, microcoils with an inner diameter of 0.18 mm (Sawane Spring Co., Ltd.) were wrapped around the common carotid arteries on both sides. Blood flow changes were confirmed by laser Doppler flowmetry in the anterior regions of brains (data not shown). A lentivirus encoding CSII-EF-Venus was stereotactically injected into the V-SVZ to label new neurons. To analyze the cell distribution in the RMS and OB, mice were fixed at 5 dpi when glial activation is reported not to occur (Shibata et al., 2004 [DOI](#)).

Two-photon imaging

As described previously (Sawada et al., 2011 [DOI](#)), thinned-skull surgery was performed on wild-type mice for comparisons between RBC flow and endomucin expression in identical vessels and on *Dcx-EGFP* and *VGAT-Venus* mice for observation of neuronal migration and maturation, respectively. Blood vessels were visualized by intravenous injection of Rhodamine-B dextran (D1841, Invitrogen) or Fluorescein dextran (D1823, Invitrogen). Mice were anesthetized by inhalation of isoflurane. The heads were immobilized with ear bars on a stereotactic stage before surgery. The skull over the OB was carefully thinned with a high-speed drill (MINITER Co., Ltd.) and a surgical blade (Fine Science Tools). The thinned-skull window was observed under a two-photon laser scanning microscope (Nikon) and mode-locked system at 950 nm (Mai Tai HP, Spectra Physics) with a 25× water-immersion objective. Neuronal migration was recorded by identification of EGFP+ cells that changed their positions during the imaging period in the whole visible imaging field. During imaging, mice were anesthetized by intraperitoneal administration of a mixture of medetomidine (Meiji Seika Pharma Co., Ltd.), midazolam (SANDOZ), and butorphanol (Meiji Seika Pharma Co., Ltd.) (0.75 mg/kg, 4 mg/kg, and 5 mg/kg, respectively) and kept on a heating pad for maintenance of the body temperature at 37°C. Image stacks (2048 × 2048 pixels, 0.25 μm per pixel, 2-μm z-step size) were acquired every 30–60 min during 3 hours.

RBC flow was recorded by serial line-scans as previously reported (Kleinfeld et al., 1998 [DOI](#)). Line-shaped regions of interest were drawn along the longitudinal axis of each blood vessel. The RBC flow/s was calculated from repetitive scans obtained during 10 s at the beginning of the neuronal-migration recording. The median RBC flow velocity (38.4 RBCs/s) was used as a criterion for classification of vessels with different blood flows, whose distribution is not normal. Neuronal maturation was recorded as previously described with modifications (Sawada et al., 2011 [DOI](#)). After mice were anesthetized by isoflurane inhalation, the thinned-skull window was observed to record positions of Venus+ cells and RBC flow from each vessel in square-shaped regions (512 × 512 pixels, 0.5 μm per pixel, 2-μm z-step size). The same region of the GL was observed with an interval of 21 days. Stationary cells were defined as cells that were observed in the same position at both time points. Maturation and cell death in the GL were identified as addition and loss of Venus+ cells at the second time point. Data analysis was performed using NIS Element software (Nikon).

Photothrombotic clot formation

Photothrombotic clot formation was performed as previously reported with modifications (Schaffer et al., 2006 [DOI](#)). After identification of a blood vessel close to migrating neurons, an upstream vessel fragment was surrounded by a rectangular region of interest. Mice were intravenously injected with 20 mg/ml rose bengal (330000, Sigma-Aldrich) in PBS at concentration

0.05 mg/g. Immediately after injection, a selected fragment was irradiated using a two-photon laser at 950 nm. Irradiation by a 100-mW laser was performed for 5~10 seconds until the clot formed. The inner space of the vessel was equally irradiated by continuous movement of the imaging area. Ten minutes after introduction of rose bengal, the RBC flow of a target vessel was recorded to confirm blood flow inhibition. The behavior of migrating neurons was observed for 3 hours after clot formation and compared with that before photothrombosis. As a control experiment, vessels without rose Bengal injection were irradiated with a two-photon laser. Samples were excluded if bleeding occurred or clots were dissolved during observation.

Transmission electron microscopy

Transmission electron microscopy analysis was performed as previously described with modifications (Matsumoto et al., 2019 [↗](#); Sawada et al., 2018 [↗](#)). Brain were fixed in 2.5% glutaraldehyde and 2% PFA in 0.1 M PB (pH 7.4) at 4°C, postfixed with 2% OsO₄ in the same buffer at 4°C, dehydrated in a graded ethanol series, placed in propyleneoxide, and embedded in Durcupan resin for 72 hours at 60°C to ensure polymerization. Ultra-thin sections (60–70 nm) were cut using an ULTRACUT-E (Reichert-Jung) with a diamond knife, stained with 2% uranyl acetate in distilled water for 15 min, and stained with modified Sato's lead solution for 5 min. Sections were analyzed with a transmission electron microscope (JEM-1011J; JEOL, Tokyo, Japan). Migratory neurons was identified as cells with an elongated cell body, a dark cytoplasm with many free ribosomes, and an electron-dense nucleus with multiple nucleoli (Doetsch et al., 1997 [↗](#); Matsumoto et al., 2019 [↗](#); Sawada et al., 2018 [↗](#)).

Fluorescence *in situ* hybridization (RNA scope)

Fluorescence in situ hybridization was performed as previously described with modifications (Miyamoto et al., 2025 [↗](#)) using the RNAscope multiplex assay (Advanced Cell Diagnostics), according to the manufacturer's protocol (Wang et al., 2012 [↗](#)). After fixation with 4% PFA as described above, brains were postfixed for 4 hours at 4 °C, incubated in 15% and 30% sucrose solutions in PBS, embedded in OCT compound (Sakura Finetek Japan), and frozen using dry ice and isopentane. Serial brain sections (10 µm) were cut using a cryostat (CryoStar NX70, Eppredia). The sections were fixed again in 4% PFA at RT for 30 min, dehydrated in ethanol, treated with 1% hydrogen peroxide at RT for 30 min, and then boiled in RNAscope Target Retrieval Reagent (Advanced Cell Diagnostics) at 98 °C for 15 min. Hybridization was performed using a probe targeting *Ghsr1a* mRNA (Mm-Ghsr-O2-C1, #241998, Advanced Cell Diagnostics) for 2 hours at 40 °C. Following this, immunohistochemistry with rabbit anti-Dcx antibody and Hoechst staining was conducted.

Protein labeling

For observation of ghrelin transcytosis across the vascular wall, recombinant octanoylated ghrelin (Human, sc-364689, Santa Cruz Biotechnology) was fluorescently labeled with Atto 647N NHS ester (18373, Sigma-Aldrich) as previously described with modifications (Yang et al., 2020 [↗](#)).

Ghrelin was dissolved in 0.1 M bicarbonate buffer (pH 8.3) at 2 mg/ml and reacted with Atto 647N NHS ester for 60 min at RT. After reactions, free label was removed by gel permeation chromatography with PD MiniTrapTM G-25 columns (28918007, Cytiva). Mice were fixed as described above at 1 hour after intravenous injection of fluorescently labeled ghrelin (0.02 mg / 30g). Fluorescence signal intensity was measured by ZEN software. The average intensity among all vessels in each mouse was normalized to 1.0. To quantify signal gradients in extra-vessel regions, two circular ROIs (10 µm in diameter) were defined as illustrated in [Figure 3E](#) [↗](#). The centers of the ROIs were aligned along the long axis of each vessel. Mean fluorescence intensities were measured in Area I (0–10 µm from the abluminal surface) and Area II (10–20 µm away), and fold changes (Area I / Area II) were calculated for each vessel segment.

V-SVZ culture experiments

The V-SVZ was dissected from P0–P1 mice, cut into blocks, and embedded in 60% Matrigel (BD Biosciences) in L-15 medium (Gibco). Cell aggregates were cultured in Neurobasal medium containing 2% NeuroBrew-21 (Invitrogen), 2 mM L-glutamine (Gibco), and 50 U/ml penicillin-streptomycin (Gibco) at 37°C in a 5% incubation system (Tokai Hit). For migration distance analysis, octanoylated ghrelin (Human, Rat, 1-10, Peptides International) was added to the medium at a final concentration of 100 nM at 24 hours post-embedding (hpe). Then, cell aggregates were fixed in 4% PFA/0.1M PB at 48 hpe. For immunocytochemistry, aggregates were incubated in blocking solution for 30 min at RT, treated with the primary antibodies in blocking solution for 24 hours at 4°C, and treated with AlexaFluor-secondary antibodies (1:1000) in the same solution for 2 hours at RT. The migration distance was analyzed in three independent cultures prepared on different days.

Viral vectors and plasmids

The pCSII lentiviral expression vectors were provided by Dr. Hiroyuki Miyoshi (RIKEN Tsukuba BioResource Center). The *lacZ*- and *Ghr1a*-KD plasmids were generated as previously described (Ota et al., 2014 [DOI](#); Sawada et al., 2018 [DOI](#)). The target sequences of *lacZ* mRNA and mouse *Ghr1a* mRNA (Invitrogen) were inserted into modified Block-iT Poll II miR RNAi expression vectors containing EmGFP or DsRed-Express (Invitrogen). The Gateway System (Invitrogen) was used to generate pCSII-EF-Venus, pCSII-EF-Ghr1a-IRES-Venus, pCSII-EF-EmGFP-express-miR-lacZ, pCSII-EF-DsRed-express-miR-Ghr1a, pCAGGS-DsRed-express-miR-lacZ, and pCAGGS-DsRed-express-miR-Ghr1a. For lentivirus production, the packaging vectors (pCAG-HIVgp, pCMV-VSV-G-RSV-Rev) and pCSII viral vectors were co-transfected into HEK-293T cells to generate lentivirus particles. Then, the culture supernatants were concentrated by centrifugation at 8,000 rpm for 16 hours at 4°C in an MX-307 refrigerated microcentrifuge (Tomy).

Ghr1a KD experiments

The following sequence was inserted into siRNA expression vectors for targeting *Ghr1a* mRNA: TGCTGAAGATGAGCAGATCGGAGAAGGTTTGGCCACTGACTGACCTTCTCCGCTGCTCA TCTTCAGG. For confirming efficacy of *Ghr1a* KD, pCSII-EF-Ghr1a-IRES-Venus and/or pCSII-EF-DsRed-express-miR-Ghr1a were co-transfected in HEK-293T cells. Venus signal was not observed in DsRed+ *Ghr1a*-KD cells (data not shown).

For *in vitro* experiments, the dissected V-SVZ was dissociated with trypsin-EDTA (Invitrogen). The pCS2-EGFP-UtrCH was provided by Dr William Bement (University of Wisconsin-Madison) and Dr David J. Solecki (St. Jude Children's Research Hospital). The cells were washed in L-15 medium with 40 µg/ml DNaseI (Roche) and transfected with 2 µg plasmid DNA using the 4D-Nucleofector (Lonza). The cells were recovered in RPMI-1640 medium (Thermo Fisher Scientific) and embedded in 60% Matrigel in L-15. After cultivation in Neurobasal medium for 48 hours, time-lapse imaging was performed using an LSM880 confocal laser scanning microscope with a 40× objective (Carl Zeiss). Time-lapse images were captured at 30-s (**Figure 4C–J** [DOI](#)) and 20-s (**Figure 4K–M** [DOI](#)) intervals. The migration distance of cultured new neurons was measured using ImageJ manual tracking tools. A migratory phase was defined as a phase in which the cell soma traveled ≥ 30 µm during 1 hour, and a resting phase was defined as a phase in which the cell soma migrated < 30 µm. ZEN software (Carl Zeiss) was used to analyze actin cup formation. Actin cups were defined as over 4-µm-long continuous EGFP-UtrCH signals that were 1.3 times brighter than those in other cell soma regions.

For *in vivo* experiments, the lentiviral solution was stereotactically injected into the V-SVZ and OB core of adult mice (V-SVZ: 1.0 mm anterior, 1.1 mm lateral to bregma, and 1.6–2.0 mm deep) (OB core: 4.6 mm anterior, 0.9 mm lateral to bregma, and 0.6–0.9 mm deep). The proportions of

EmGFP+/DsRed- and EmGFP-/DsRed+ cells among total labeled Dcx+ new neurons in the RMS and OB were analyzed using ZEN software (Carl Zeiss). Double-positive cells (EmGFP+/DsRed+) were excluded from all analyses to avoid confounding effects of coinfection.

Calorie restriction was performed as previously reported (Hornsby et al., 2016). In migration analysis, calorie-restricted animals were fed 70% of the total food consumed by animals fed ad libitum daily for the last 5 days prior to fixation at 8 dpi. In maturation analysis, calorie restriction was performed from Day 3 to Day 8 after BrdU administration at Day 0, following fixation at Day 15.

Statistics

Statistical analysis was performed using EZR software (Kanda, 2013). The normality of the data was analyzed using a Kolmogorov-Smirnov test or Shapiro-Wilk test. The equality of variance was analyzed using an *F* test. Comparisons of data between two groups were performed with unpaired/paired *t*-tests or Welch's *t*-test for normally distributed data and by Mann-Whitney *U*-tests/Wilcoxon signed-rank tests for abnormally distributed data. Comparisons among multiple groups were performed by one-way analysis of variance (ANOVA)/one-way repeated measures ANOVA/Kruskal-Wallis tests followed by a post hoc Tukey-Kramer test, Bonferroni test, or Steel-Dwass test. Numerical data are presented as the means $\pm$ standard error of the mean. A *p* value < 0.05 was considered statistically significant. Significance is indicated in graphs as follows: **p* < 0.05 , ***p* < 0.01 , ****p* < 0.005 , n.s., not significant.

Supplementary figures

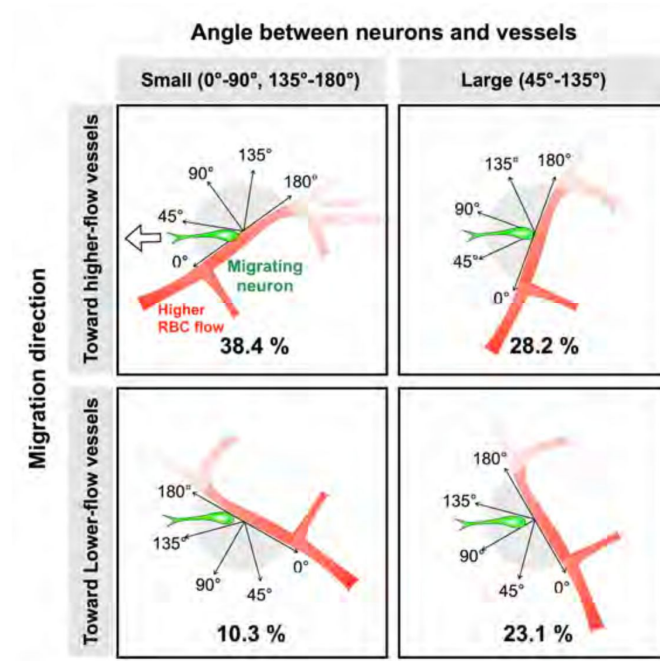


Figure S1

Directional migration of new neurons relative to local blood flow.

New neuron-vessel interactions were categorized into four groups according to the angle between the migration direction and vessel axis (small: 0° - 45° , 135° - 180° ; large: 45° - 135°), and whether the new neurons were migrating toward or away from the direction of higher red blood cell (RBC) flow. Percentages represent the proportions of total interactions ($n = 39$ from 26 mice).

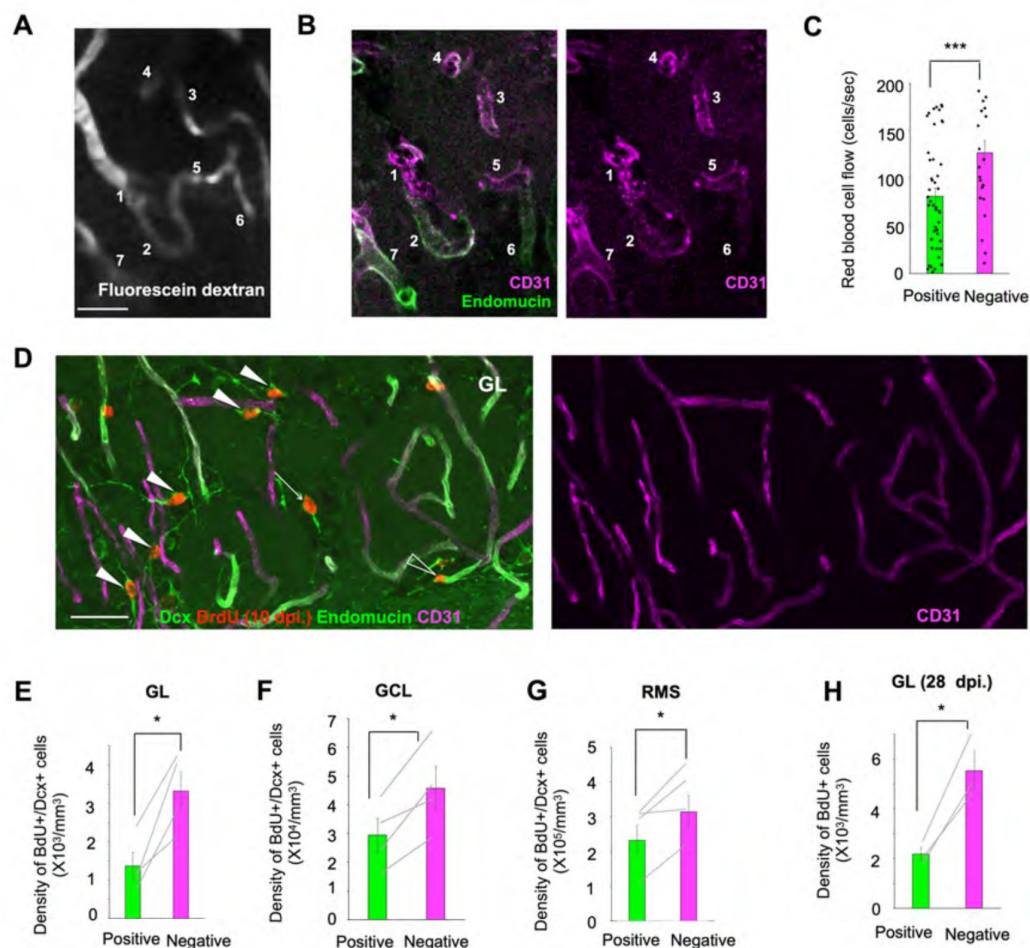


Figure S2

New neurons migrate along endomucin-negative vessels.

(A, B) Representative images of vasculature in the glomerular layer (GL) of the olfactory bulb (OB). Red blood cell (RBC) flow was recorded in a live animal (A), followed by immunostaining of endomucin/CD31 in a fixed brain section (B). Identical vessels are indicated by different numbers (endomucin-positive; 2, 6, 7, endomucin-negative; 1, 3, 4, 5).

(C) Average RBC flow in endomucin-positive and endomucin-negative vessels (Mann-Whitney U-test; endomucin-negative, 24 vessels, endomucin-positive, 46 vessels).

(D) Fluorescent image of new neurons distributed in the vasculature in the GL. BrdU+/Dcx+ cells are shown in the perivascular region of endomucin-negative vessels (white arrowheads), endomucin-positive vessels (clear arrowhead), and distant from vessels (arrow). CD31 (magenta), endomucin (green, tube-like structures), Dcx (green), BrdU (red).

(E–G) Density of BrdU+/Dcx+ cells in the vicinity of endomucin-positive and endomucin-negative vessels in the GL (E), granule cell layer (F), and rostral migratory stream (OB core) (G) (paired *t*-test; 4 mice).

(H) Density of BrdU+ mature neurons at 28 dpi. in the vicinity of endomucin-positive and endomucin-negative vessels in the GL (paired *t*-test; 4 mice). Data are presented as the means $\pm$ SEM. Scale bars: 20 μm (A, B) ; 20 μm (D).

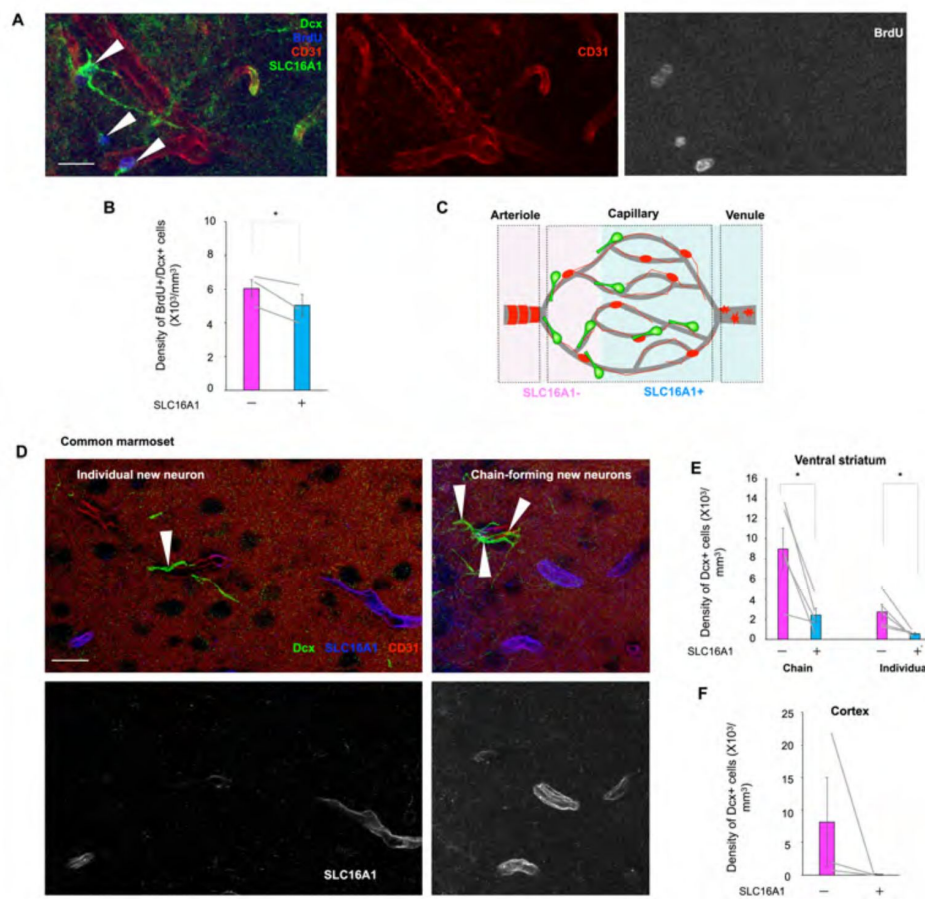


Figure S3

New neurons exhibit a preference for arteriole-side vessels.

(A) Fluorescent image of immunostained tissue sections from the glomerular layer of the olfactory bulb. Dcx (green), BrdU (deep blue), CD31 (red), SLC16A1 (green, tube-like structures).

(B) Density of perivascular new neurons in the vicinity of SLC16A1-positive and SLC16A1-negative vessels (paired *t*-test; 4 mice).

(C) Schematic illustration of the distribution of new neurons and vessel identification.

(D) Fluorescent images of the ventral striatum from a 4-month-old common marmoset. Immature neurons are indicated by solid arrowheads. Dcx (green), SLC16A1 (deep blue), CD31 (red).

(E, F) Density of BrdU+/Dcx+ cells in the vicinity of SLC16A1-positive and SLC16A1-negative vessels in the ventral striatum (E) (paired *t*-test; 5 animals) and in the neocortex (F). Data are presented as mean $\pm$ SEM. Scale bars: 20 μ m (A) ; 20 μ m (D).

Figure S4

BCAS affects cell proliferation and survival in the VSVZ and RMS.

(A, B) Fluorescent images of immunostained RMS tissue sections showing Ki67 (green) (A) and cleaved caspase-3 (green) (B). (C, D) Density of Ki67+ cells (C) and cleaved caspase-3+ cells (D) in the RMS and VSVZ (unpaired t-test; Sham, 7 mice, BCAS, 7 mice). Data are presented as mean $\pm$ SEM. Scale bars: 10 μ m (A, B).

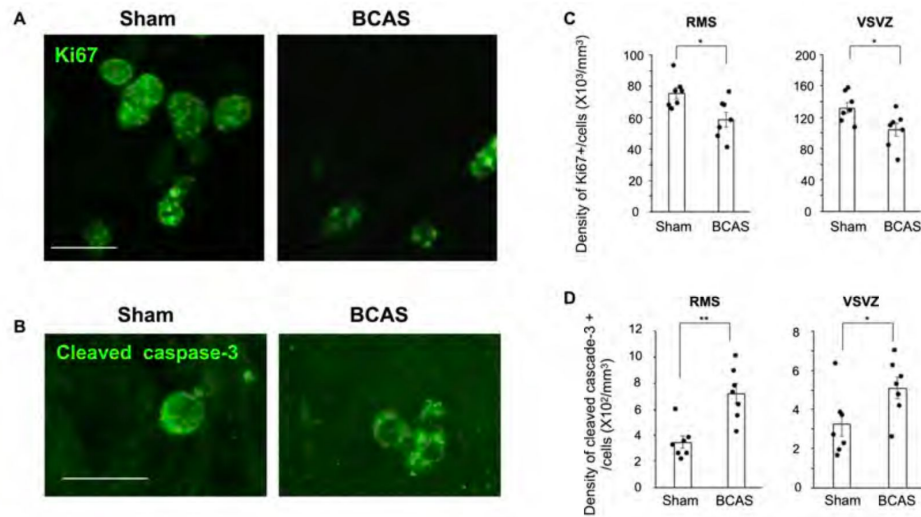
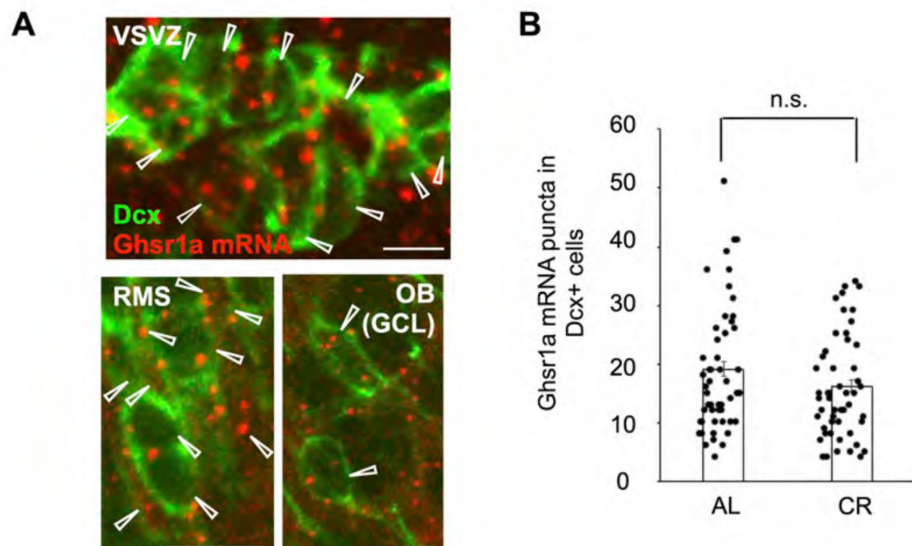


Figure S5

New neurons express *Ghsr1a* mRNA in the adult brain.

(A) Fluorescent images from the V-SVZ, RMS and OB showing Dcx (green) and *Ghsr1a* mRNA (red). Arrowheads indicate Dcx+ cells with *Ghsr1a* mRNA puncta.

(B) Quantification of *Ghsr1a* mRNA puncta per Dcx+ cell in the OB of the AL and CR mice (AL, 68 cells from 3 mice, CR, 68 cells from 3 mice). Data are presented as mean $\pm$ SEM. Scale bars: 10 μ m



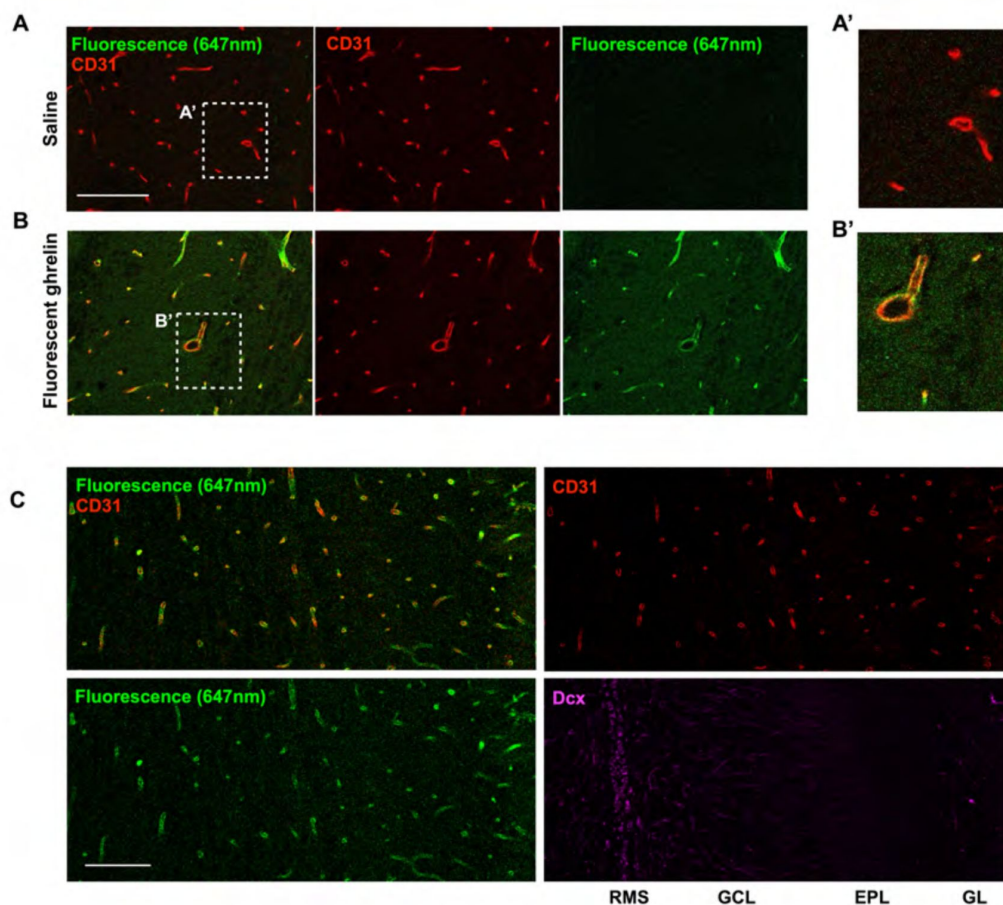


Figure S6

Blood-derived ghrelin enters the RMS and OB.

(A, B) Representative images of the OB from mice with saline injection (A) and with fluorescent ghrelin injection (B). High-magnification images are shown in (A'), (B'). It was found that the experimental process did not affect the brightness of sections. CD31 (red), fluorescence (647 nm) (green).

(C) Representative images of the OB section showing ghrelin fluorescence (647 nm, green), CD31 (red), and Dcx (magenta), covering the RMS, GCL, EPL, and GL. Scale bars: 100 μ m (A,B,C).

Figure S7

Ghrelin signaling is required for neuronal migration in the adult OB.

(A) Experimental schematic.

(B) Proportion of labeled cells in the GL at 5 dpi. (unpaired *t*-test; Control, 4 mice; *Ghsr1a*-KD, 5 mice). Data are presented as mean $\pm$ SEM.

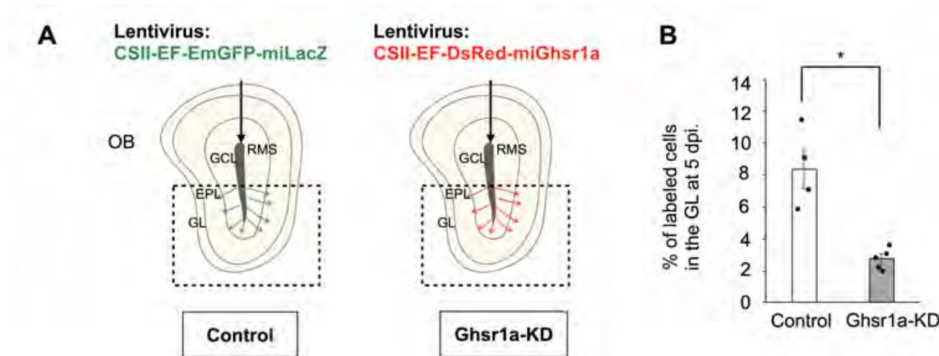
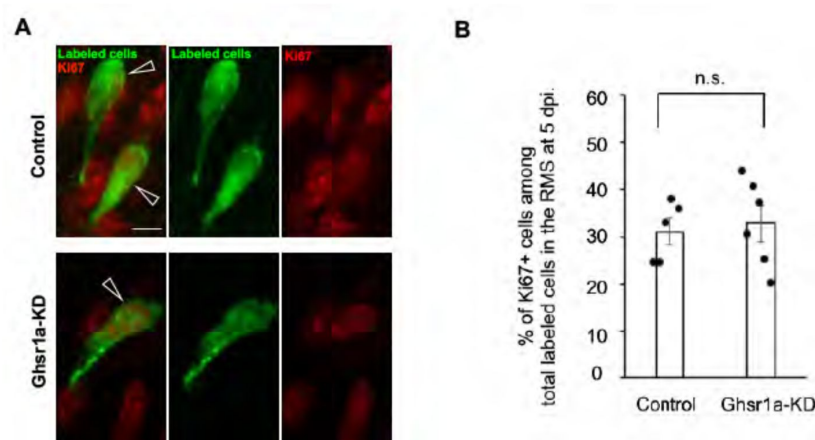


Figure S8

***Ghsr1a*-KD does not affect cell proliferation of new neurons in the RMS.**

(A) Fluorescent image of immunostained RMS tissue sections showing Ki67 (red) and labeled cells (green). Ki67+ labeled cells are indicated by clear arrowheads.

(B) Quantification of Ki67+ cells as a percentage of the total labeled cells in the RMS at 5 dpi (Control, 5 mice; *Ghsr1a*-KD, 6 mice). Data are presented as mean $\pm$ SEM. Scale bars: 5 μ m (A).



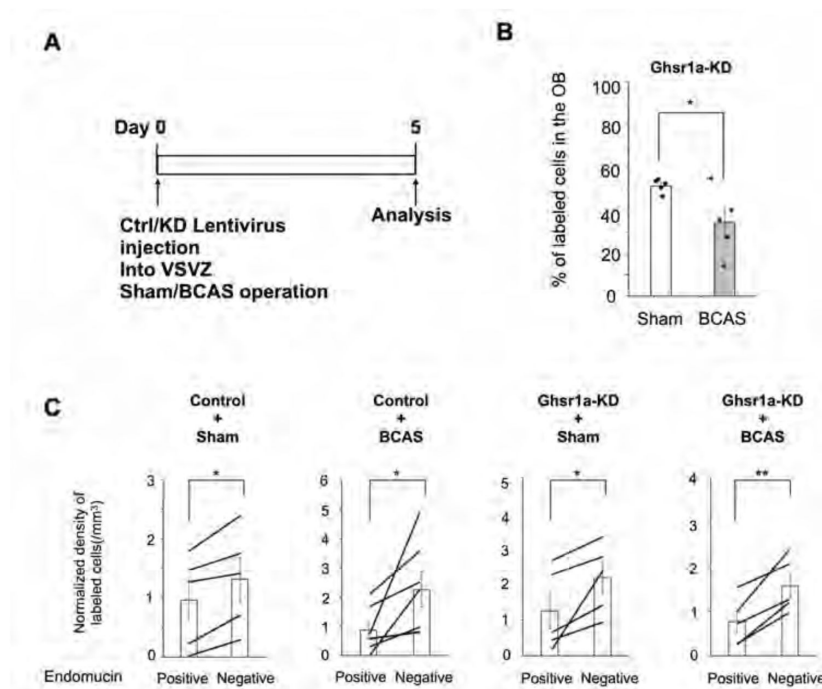


Figure S9

***Ghsr1a*-KD does not affect the influence of BCAS on neuronal migration.**

(A) Experimental timeline.

(B) Proportion of *Ghsr1a*-KD cells in the OB at 5 dpi. (unpaired *t*-test; Control, 5 mice; *Ghsr1a*-KD, 5 mice).

(C) Normalized density of labeled cells in perivascular regions of the RMS (paired *t*-test; Control+Sham, 5 mice; Control+BCAS, 6 mice; *Ghsr1a*-KD+Sham, 5 mice; *Ghsr1a*-KD+BCAS, 5 mice). Data are presented as mean $\pm$ SEM.

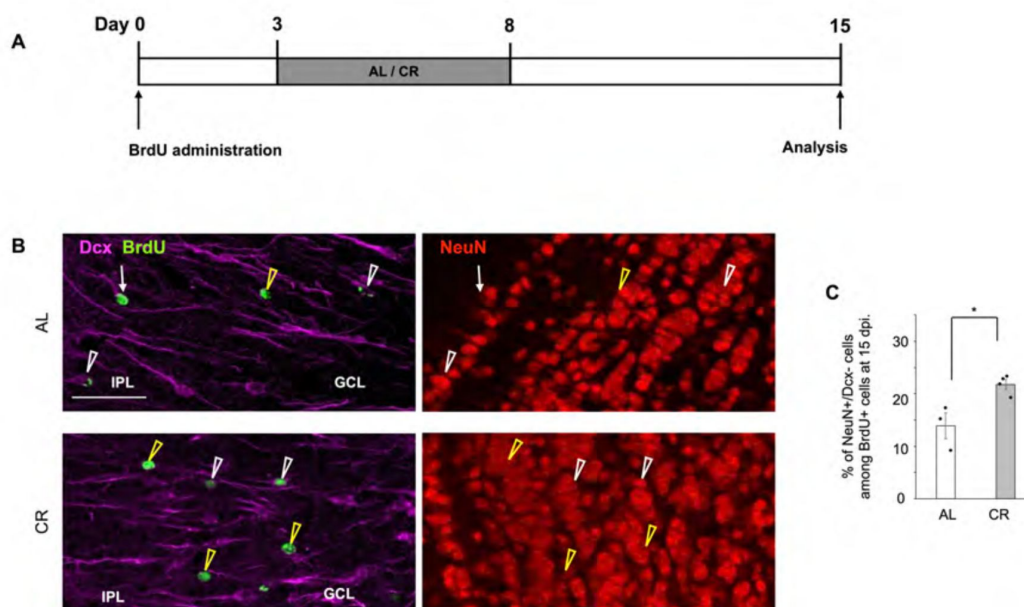


Figure S10

Calorie restriction promotes neuronal maturation in the OB.

(A) Experimental scheme.

(B) Fluorescent images of new neurons in the OB. BrdU (green), NeuN (red), Dcx (magenta). A NeuN-/Dcx+ cell (white arrow), NeuN+/Dcx+ cells (white arrowheads), NeuN+/Dcx-cells (yellow arrowheads).

(C) Proportion of NeuN+/Dcx-cells among total BrdU+ cells in the OB (KD, unpaired *t*-test; AL, 3 mice, CR, 4 mice). IPL (internal plexiform layer), GCL (granule cell layer). Data are presented as mean $\pm$ SEM. Scale bars: 50 μ m (B).

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Additional information

Data and materials availability

All data are available in the main text or in the supplementary materials.

Author contributions

TO, AS, MS, ST, YH, KY, JN, HK, TS, HI, VHP, YM, ME, JMGV, JN, and KS performed experiments and analyzed the data. TO and KS wrote the manuscript.

Additional files

Supplemental Movie 1. A three-dimensional image from the rostral migratory stream of a Dcx-EGFP (green) mouse infused with RITC-Dex-GMA (red). [🔗](#)

Supplemental Movie 1'. High-magnification view extracted from Supplemental Movie S1. Arrows indicate EGFP+ cells interacting with blood vessels. [🔗](#)

Supplemental Movie 2. A three-dimensional image from the rostral migratory stream of a Dcx-EGFP/Flt1-DsRed mouse. EGFP (green), DsRed (red). [🔗](#)

Supplemental Movie 3. A three-dimensional, high-magnification image of chain-forming new neurons in the rostral migratory stream. Dcx-EGFP (green), Flt1-DsRed (red). Arrows indicate EGFP+ cells interacting with blood vessels. [🔗](#)

Supplemental Movie 4. A three-dimensional image of immature neurons leaving the ventral ventricular-subventricular zone in a 1-month-old common marmoset infused with RITC-Dex-GMA (red). Dcx (green). Arrows indicate Dcx+ cells interacting with blood vessels. [🔗](#)

Supplemental Movie 5. A three-dimensional image of an immature neuron from the ventral striatum in a 1-month-old common marmoset infused with RITC-Dex-GMA (red). Dcx (green). Arrows indicate Dcx+ cells interacting with blood vessels. [🔗](#)

Supplemental Movie 6. Actin cup imaging in control cells. EGFP-UtrCH (green). [🔗](#)

Supplemental Movie 7. Actin cup imaging in *ghsr1a*-knockdown cells. EGFP-UtrCH (green). [🔗](#)

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Reviewer #1 (Public review):

Summary:

This study provides compelling evidence suggesting that ghrelin, a molecule released in the surrounding of the major adult brain neurogenic niche (V-SVZ) by blood vessels with high blood flow controls the migration of newborn interneurons towards the olfactory bulbs.

Strengths:

This study is a tour de force as it provides a solid set of data obtained by time lapse recordings in vivo. The data demonstrate that the migration and guidance of newborn neurons relies on factors released by selective type of blood vessels.

Weaknesses:

Some intermediate conclusions are weak and may be reinforced by additional experiments.

Comments on revisions: The manuscript has improved.

<https://doi.org/10.7554/eLife.99502.2.sa1>

Author response:

The following is the authors' response to the original reviews

Reviewer #1 (Public review):*Summary:*

This study provides compelling evidence suggesting that ghrelin, a molecule released in the surroundings of the major adult brain neurogenic niche (V-SVZ) by blood vessels with high blood flow, controls the migration of newborn interneurons towards the olfactory bulbs.

Strengths:

This study is a tour de force as it provides a solid set of data obtained by time-lapse recordings in vivo. The data demonstrate that the migration and guidance of newborn neurons rely on factors released by selective types of blood vessels.

Weaknesses:

Some intermediate conclusions are weak and may be reinforced by additional experiments.

We thank the reviewer for the thoughtful evaluation and constructive comments outlined in the “Recommendations for The Authors”. In response, we have incorporated additional data, revised relevant figures, and clarified explanations in the revised manuscript.

Reviewer #2 (Public review)*Summary:*

The authors establish a close spatial relationship between RMS neurons and blood vessels. They demonstrated that high blood flow was correlated with migratory speed. In vitro, they demonstrate that Ghrelin functions as a motogen that increases migratory speed through augmentation of actin cup formation. The authors proceed to demonstrate through the knockdown of the Ghrelin receptor that fewer RMS neurons reach the OB.

They show the opposite is true when the animal is fasted.

Strengths:

Compelling evidence of close association of RMS neurons with blood vessels (tissue clearing 3D), preferentially arterioles. Good use of 2-photon imaging to demonstrate migratory speed and its correlation with blood flow. In vitro analysis of Ghrelin administration to cultured RMS neurons, actin visualization, Ghrelin1KD, is solid and compelling.

We sincerely thank the reviewer for the encouraging comments and helpful suggestions. As noted, our original manuscript lacked sufficient in vivo evidence connecting blood flow with ghrelin signaling. To address this, we have added new data and revised the explanations throughout the manuscript as described below.

Weaknesses:

(1) Novelty of findings attenuated due to prior work, especially Li et al., Experimental Neurology 2014. Here, the authors demonstrated that Ghrelin enhances migration in adultborn neurons in the SVZ and RMS.

We agree with the reviewer that the idea that ghrelin enhances migration of new neurons is not entirely novel. The study by Li et al. (2014) provided critical insights that guided our investigation into ghrelin as a blood-derived factor promoting neuronal migration. However, our study expands on this by demonstrating that ghrelin directly stimulates migration via GHSR1a in cultured new neurons, and we further identified the cellular and cytoskeletal mechanisms involved. Specifically, we showed that ghrelin enhances somal translocation by activating actin dynamics at the rear of the cell soma. We have revised the Results and Discussion sections accordingly to emphasize these novel aspects as follows:

“A previous study demonstrated that the migration of V-SVZ-derived new neurons was attenuated in ghrelin knockout mice (Li et al., 2014). In our study, we found that the migration of cultured new neurons was enhanced by the application of ghrelin to the culture medium, and this effect was abolished by Ghsr1a knockdown (KD). These findings suggest that ghrelin directly stimulates neuronal migration through its receptor, GHSR1a, on new neurons. A previous study showed that GHSR1a is expressed in various regions of the brain (Zigman et al., 2006). In our experiments, new neuron-specific KD of Ghsr1a indicated that ghrelin signaling acts in a cell-autonomous manner to regulate neuronal migration.” (Discussion, page 13, lines 10–18)

“Furthermore, we identified the cellular and cytoskeletal mechanisms underlying this effect on migration. The results indicate that ghrelin enhances somal translocation during migration by activating actin cytoskeletal dynamics at the rear of the neuronal soma.” (Discussion, page 13, lines 24–26)

(2) The evidence for blood delivery of Ghrelin is not very convincing. Fluorescently-labeled Ghrelin appears to be found throughout the brain parenchyma, irrespective of the distance from vessels. It is also not clear from the data whether there is a link between increased blood flow and Ghrelin delivery.

We agree that the correlation between blood flow and ghrelin transcytosis is not very convincing in our study. As the reviewer pointed out, Figure 3A gives the impression that fluorescent-labeled ghrelin is uniformly distributed throughout the brain parenchyma. However, high-magnification images newly added in Figure 3 show that some, but not all, vessels have particularly strong fluorescent signals in the parenchymal area adjacent to the abluminal side of vascular endothelial cells, visualized by CD31 immunostaining (Feng et al., 2004) (Figure 3A', A"). To quantify these observations, we defined two regions: Area I (perivascular area), within 10 μ m of the abluminal surface of CD31-positive endothelium; and Area II (distant area), located 10–20 μ m away (Figure 3E). Of note, Area I corresponds to the perivascular region where new neurons are frequently observed (Figure 1).

Importantly, we found strong ghrelin signals in vascular endothelial cells of endomucin-negative high-flow vessels (Figure 3C, D). This suggests that transcytosis of blood-derived ghrelin may occur more frequently in high-flow vessels due to increased endocytosis at the endothelium. To test this, we quantified signal gradients in the extra-vessel regions as fold changes (Area I / Area II), as illustrated in Figure 3E. The proportion of vessel segments with >1.5-fold increases was significantly higher in endomucin-negative vessels than in endomucin-positive ones (Figure 3F). Furthermore, vessels with >2-fold increases were observed exclusively in the endomucin-negative group (6.48% $\pm$ 1.18%).

These data suggest that, in high-flow vessels, blood-derived ghrelin accumulates more in the immediate perivascular region than in areas further away. This supports the possibility that elevated blood flow delivers a larger amount of ghrelin to the vascular endothelium, enhancing its transcytosis into adjacent brain parenchyma. This mechanism may underlie the preferential migration of new neurons along perivascular regions with high blood flow, as

shown in Figure 1. We have incorporated this new data in Figure 3 and corresponding explanations into the Results, Figure legend and Methods

(3) The in vivo link between Ghnr1KD and migratory speed is not established. Given the strong work to open the study on blood flow and migratory speed and the in vitro evidence that migratory speed is augmented by Ghrelin, the paper would be much stronger with direct measurement of migration speed upon Ghnr1KD. Indeed, blood flow should also be measured in this experiment since it would address concerns in 2. If blood flow and ghrelin delivery are linked, one would expect that Ghnr1KD neurons would not exhibit increased migratory speed when associated with slow or fast blood flow vessels.

In Figure 3, we showed that ghrelin transcytosis occurs preferentially in high-flow vessels, suggesting a role for ghrelin in mediating the effects of blood flow on neuronal migration. However, whether this dependence is solely attributable to ghrelin signaling remains unclear.

To address this, we tested whether Ghnr1a-KD modifies the impact of reduced blood flow on neuronal migration by combining Ghnr1a-KD with bilateral common carotid artery stenosis (BCAS), a chronic cerebral hypoperfusion model (Figure S9A). We found that BCAS decreased the percentage of Ghnr1a-KD new neurons reaching the OB, similar to the effect seen in control neurons (Figure S9B, see also Figure 2A–C). This suggests that blood flow influences neuronal migration even under Ghnr1a-KD conditions.

Furthermore, we analyzed the distribution of Ghnr1a-KD neurons with respect to vessel flow characteristics. Even under Ghnr1a-KD, a higher proportion of new neurons were located in the area of endomucin-negative (high-flow) vessels compared with endomucin-positive (low-flow) vessels (Figure S9C), indicating that Ghnr1a-KD does not abolish the preferential association of migrating neurons with high flow vessels. These findings suggest that although ghrelin signaling contributes to blood flow-dependent migration, it is not the sole factor. Other blood-derived signals may also mediate this effect. We have included these new data in Figure S9 and updated the corresponding sections in the Results

Reviewer #1 (Recommendations for the authors) :

Major

Page 6, Line 13. Please provide in the result section some explanation about how photothrombotic clot is induced.

We added the following explanation to the Results section to clarify the method used to induce photothrombotic clot formation.

“For clot formation, a restricted area of selected vessels was irradiated by a two-photon laser immediately after intravenous injection of rose bengal.” (Results, Page 7, lines 27–28)

Page 6, Line 18. The authors use the marmoset as an additional experimental model. Here, V-SVZ-derived newborn neurons migrate in other brain regions as compared to rodents. Please provide a clear rationale for moving from rodents to "common marmosets" as an experiment model. And why use marmosets only for this set of experiments?

We clarified the rationale for using common marmosets in addition to mice as follows:

“Because blood vessel-guided neuronal migration in the adult brain is a conserved phenomenon across species (Kishimoto et al., 2011; Akter et al., 2021; Shvedov et al., 2024), we hypothesized that blood flow may also influence neuronal migration in other brain regions of

primates. The neocortex, which supports higher-order brain functions and has undergone evolutionary expansion in primates, was selected as a target region. In common marmosets, but not in mice, V-SVZ-derived new neurons migrate toward the neocortex and ventral striatum (Akter et al., 2021) (Supplemental Movies S4 and S5).” (Results, Page 6, lines 19–25)

Figure 2B. The experimental setup is possibly problematic as the lentiviral tracing measurement does not take into consideration the rate of neurogenesis or newborn neuron survival. Can authors assess the rate of proliferation and survival in the VSVZ/RMS upon BCAS to decipher whether the reduced number of cells observed in the OB only results from migration changes? (comparable remark stands for Figure 5)

To evaluate whether the reduction in the number of new neurons observed in the OB after BCAS (Figure 2B, C) is due solely to impaired migration, we assessed cell proliferation and survival in the V-SVZ and RMS. Specifically, we quantified the density of Ki67+ proliferating cells and cleaved caspase-3+ apoptotic cells in the sham and BCAS groups. BCAS significantly decreased cell proliferation and increased cell death in both the V-SVZ and RMS (Figure S4), suggesting that reduced neurogenesis and/or survival may contribute to the decreased neuronal distribution in the OB.

Although we cannot exclude the possibility that changes in cell proliferation or survival contributed to this effect, our photothrombotic clot formation experiments are better suited to directly examine how acute reduction in blood flow affects neuronal migration. These experiments allowed us to measure the migration speed of new neurons shortly after inducing localized blood flow inhibition. We found that clot formation significantly reduced the migration speed of new neurons (Figure 2E, H), indicating that blood flow changes directly impair neuronal migration in the adult brain.

We have included these new data in Figure S4 and updated the corresponding text in the Results, Discussion, Figure legend, and Methods as follows:

Figure 3. About ghrelin signaling. It is unclear whether its transcytosis occurs in endomucin-negative because of the high bloodstream flow. How can this be explained? What happens upon BCAS, is there still a close relation between ghrelin transcytosis, blood flow, and neuron migration?

As correctly noted, our initial explanation and data did not provide sufficient evidence that higher blood flow delivers a larger amount of ghrelin into the brain parenchyma. We found that some vessels had particularly strong fluorescent signals in the parenchymal area adjacent to the abluminal surface of vascular endothelial cells, as visualized by CD31 immunostaining (Feng et al., 2004) (Figure 3A', A"). On the basis of our observation that strong fluorescent signals were detected in vascular endothelial cells of endomucin-negative (high-flow) vessels (Figure 3C, D), we hypothesized that ghrelin transcytosis may occur more frequently in high-flow vessels due to increased endocytosis at the vessel endothelium.

To test this hypothesis, we quantified signal gradients in the extra-vessel regions by calculating fold changes in fluorescent intensity between two zones: Area I (0–10 μ m from the abluminal surface of the endothelium) and Area II (10–20 μ m away), as illustrated in Figure 3E. Area I corresponds to the perivascular region where new neurons are frequently found (Figure 1). We found that the proportion of vessel segments with >1.5-fold signal increase in Area I relative to Area II was significantly higher in endomucin-negative vessels than endomucin-positive ones (Figure 3F). Furthermore, vessel segments with >2-fold increases were observed exclusively in the endomucin-negative group (6.48% $\pm$ 1.18%). These results support the idea that higher blood flow increases the amount of ghrelin that reaches the luminal surface of vascular endothelial cells, thereby increasing the possibility of ghrelin transcytosis into the brain parenchyma.

We also examined whether blood flow inhibition—induced by BCAS or photothrombotic clot formation—affects the relationship between ghrelin transcytosis, blood flow, and neuronal migration. The above results suggest that blood flow reduction may decrease ghrelin transcytosis, thereby contributing to impaired neuronal migration. To further explore this, we analyzed the distribution of new neurons around high- versus low-flow vessels under BCAS conditions. In the BCAS group, we still observed a higher density of new neurons in the region of high-flow (endomucin-negative) vessels compared with in low-flow (endomucin-positive) ones (Figure S9C). This suggests that even under reduced blood flow, neuronal migration preferentially occurs near high-flow vessels. Taken together, these results suggest that ghrelin transcytosis, blood flow and neuronal migration are connected, and that this relationship persists under conditions of blood flow reduction.

Figure 4. Is ghrelin controlling both individual Dcx+ neuron migration as well as chain migration (cells moving more together)? This should be assessed and clarified.

How is ghrelin controlling actin dynamics in newborn migrating neurons? Since somal translocation speed and somal stride length are both modulated by ghrelin, this factor may also control MT remodeling, could that be checked?

We have revised the manuscript to better explain the role of ghrelin in both modes of neuronal migration—chain and individual. Initially, we demonstrated that ghrelin enhances the migration of new neurons in V-SVZ culture (Figure 4A, B), where these neurons migrate outward as chains, indicating that ghrelin facilitates chain migration. In subsequent in vitro experiments (Figure 4C–M), we showed that ghrelin also enhances the migration of individual neurons. To examine this in vivo, we injected Ghnr1a-KD and control lentiviruses into two different anatomical regions: the V-SVZ, where chain migration originates, and the OB core, where new neurons migrate individually. These experiments enabled us to assess the role of ghrelin signaling in each mode of migration independently. We found that ghrelin enhanced both chain migration in the RMS and individual migration in the OB. These results indicate that ghrelin signaling facilitates both forms of neuronal migration. We added the following text in the Results section:

“To assess the direct effect of ghrelin on neuronal migration, we applied recombinant ghrelin to V-SVZ cultures, in which new neurons emerge and migrate as chains (Figure 4A). Ghrelin significantly increased the migration distance of these neurons (Figure 4B), indicating enhanced chain migration. We then used super-resolution time-lapse imaging to examine individually migrating neurons with or without knockdown (KD) of growth hormone secretagogue receptor 1a (GHSR1a), a ghrelin receptor expressed in V-SVZ-derived new neurons (Li et al., 2014) (Figure 4C). Ghrelin enhanced the migration speed of control cells (lacZ-KD) cells, indicating that it also facilitates individual migration (Figure 4D).” (Results, Page 9, lines 5–12)

“Of the total labeled Dcx+ cells, the percentage of Dcx+ cells reaching the GL was significantly lower in the Ghnr1a-KD group than in the control group (Figure 5B, C), suggesting that ghrelin enhances individual radial migration of new neurons in the OB.” (Results, Page 10, lines 5–8)
 “These data indicate that ghrelin signaling facilitates both individual migration in the OB and chain migration in the RMS.” (Results, Page 10, lines 17–18)

We also added discussion on how ghrelin may regulate cytoskeletal dynamics in migrating neurons. Ghrelin signaling has been reported to control actin cytoskeletal remodeling in astrocytoma cells (Dixit et al., 2006), which led us to investigate similar effects in migrating neurons. Rac, a member of the Rho GTPase family, was shown to mediate this actin remodeling in astrocytoma migration, suggesting it may also be involved in ghrelin-induced actin cup formation in new neurons. Furthermore, because somal translocation depends not only on actin but also on microtubule dynamics (Kaneko et al., 2017), it is possible that

ghrelin influences both systems. Supporting this idea, ghrelin signaling was shown to modulate microtubule behavior via SFK-dependent phosphorylation of α -tubulin (Slomiany and Slomiany, 2017). These findings suggest that ghrelin may enhance somal translocation through coordinated regulation of both the actin and microtubule systems. We added following text in the Results and Discussion sections:

“Ghrelin signaling has been reported to regulate actin cytoskeletal dynamics in astrocytoma cells (Dixit et al., 2006), which led us to examine whether a similar mechanism operates in migrating neurons.” (Results, Page 9, lines 23–25)

“Further studies are needed to elucidate how ghrelin promotes actin cup formation in migrating neurons. Given that Rac, a Rho family GTPase, mediates actin remodeling downstream of ghrelin in astrocytoma cells (Dixit et al., 2006), it is possible that Rac may also be involved in ghrelin-induced cytoskeletal regulation in new neurons.” (Discussion, Page 13, lines 28–31)

“In addition to actin remodeling, ghrelin may regulate microtubule dynamics. Ghrelin signaling was shown to modulate microtubules via SFK-dependent phosphorylation of α -tubulin (Slomiany and Slomiany, 2017), raising the possibility that ghrelin promotes somal translocation of new neurons through coordinated regulation of both actin and microtubule networks (Kaneko et al., 2017).” (Discussion, Page 13, line 31–Page 14, line 2)

It would also be informative to provide immunolabeling of Ghnr1 in the V-SVZ / RMS / OB to have a clear picture of the expression pattern of this receptor. Newborn neurons migrate along blood vessels, which are surrounded by astrocytes that have also been reported to express Ghnr1, thus could newborn neuron migration change may also arise from activation of Ghnr1 in their surrounding astrocytes?

A previous study reported that GHSR1a is expressed in DCX+ new neurons in the RMS and OB, and in V-SVZ neural progenitor cells (Li et al., 2014). To visualize the spatial expression pattern of Ghnr1a, we performed RNAscope in situ hybridization because specific anti-GHSR1a antibodies suitable for immunohistochemistry were not available. Consistent with the previous report, we detected Ghnr1a mRNA in DCX+ new neurons in the VSVZ, RMS, and OB (Figure S5A), indicating that new neurons directly receive ghrelin signaling.

Moreover, our KD experiments demonstrated that ghrelin enhanced the migration of new neurons in a cell-autonomous manner via GHSR1a (Figure 4, 5). Nevertheless, a recent study (Stark et al., 2024) showed that GHSR1a was expressed in various cell types, including glutamatergic and GABAergic neurons, suggesting that ghrelin may also exert non-cellautonomous effects on neuronal migration. Given the presence of diverse cell types, including neurons, microglia, pericytes, and astrocytes, along the migratory route, it remains possible that GHSR1a activation in these neighboring cells contributes to the overall regulation of neuronal migration.

Figure 5. About the in vivo knockdown of Ghnr1a. The results section (page 9, line 3) mentioned that mice were either injected with one or the other construct but Figure 5 shows coincidence of GFP and dsRed positive cells. Were control and Ghnr1a shRNAs injected together into the same mouse? Could you quantify the number of cells in green (control), red (Ghnr1a KD), and yellow (both)? Won't they mostly be yellow? Have you tried injecting control and Ghnr1a separately? If yes, do you get the same result? Such analysis would be important to separate cell autonomous from noncell autonomous effects.

To minimize variability in injection conditions, we initially coinjected control and Ghnr1a-KD lentiviruses into the same mice and analyzed their migration using a paired design. As the reviewer correctly noted, some cells were coinfecting and expressed both EmGFP and DsRed

(18.7% $\pm$ 2.86% of EmGFP+ cells and 10.8% $\pm$ 0.533% of DsRed+ cells). To ensure that this overlap did not affect our analysis, we excluded EmGFP+/DsRed+ double-positive cells and focused solely on EmGFP+/DsRed- (control) and EmGFP-/DsRed+ (Ghsr1a-KD) single-positive cells.

We agree with the reviewer that coinjection could lead to reciprocal interactions between control and Ghsr1a-KD cells, potentially masking cell-autonomous effects. To address this, we performed an independent experiment in which control and Ghsr1a-KD lentiviruses were injected separately into different mice (Figure S7A), as suggested. Consistent with the results of the coinjection experiment, we found that the Ghsr1a-KD cells showed significantly reduced distribution in the GL compared with that in control cells (Figure S7B). Although we cannot exclude the possibility of a non-cell-autonomous effect of ghrelin, this result supports the conclusion that ghrelin signaling enhances neuronal migration in a cell-autonomous manner.

Who is expressing Ghsr1a, newborn neurons, and or their progenitors? The production and survival of newborn V-ZVS cells should be assessed upon knockdown of the ghrelin receptor too.

To determine whether the altered distribution of new neurons observed upon Ghsr1aKD is due to impaired migration rather than decreased cell production or survival, we examined the effects of Ghsr1a-KD on the proliferation and survival of new neurons and their progenitors, which express GHSR1a (Li et al., 2014).

We compared the proportion of cleaved caspase-3+ cells and Ki67+ cells from the total labeled cells in the V-SVZ and RMS between the control and Ghsr1a-KD groups. There was no significant difference in the proportion of cleaved caspase-3+ cells between the groups (Control: 874 cells from 5 mice; Ghsr1a-KD: 678 cells from 7 mice), suggesting that ghrelin signaling does not affect the survival of new neurons and their progenitors.

Similarly, the proportion of Ki67+ cells in the RMS did not differ significantly between the two groups (Figure S8), indicating that Ghsr1a-KD does not impair cell proliferation in the RMS. However, it remains technically difficult to evaluate whether Ghsr1a-KD affects proliferation in the VSVZ, because lentivirus injection into the VSVZ may interfere with GHSR1a expression not only in new neurons and neural progenitors, but also in other cell types known to express GHSR1a (Zigman et al., 2006). A previous study reported that ghrelin signaling promoted cell proliferation in the V-SVZ (Li et al., 2014), thus we cannot exclude the possibility that Ghsr1a-KD may affect V-SVZ proliferation.

To overcome this limitation, we assessed the effects of Ghsr1a-KD on neuronal migration using in vitro KD experiments (Figure 4C–J) and in vivo OB-core lentivirus injections (Figure 5A–C), both of which did not interfere with proliferation in the V-SVZ. These complementary approaches consistently demonstrated that Ghsr1a-KD reduces the migration speed of new neurons.

“To determine whether the altered distribution of new neurons after Ghsr1a-KD is due to impaired migration rather than changes in cell production or survival, we assessed the effects of Ghsr1aKD on the proliferation and survival of new neurons and their progenitors, which express GHSR1a (Li et al., 2014). We quantified the proportion of cleaved caspase-3+ cells and Ki67+ cells from the total labeled cells in the V-SVZ and RMS in both control and Ghsr1a-KD groups. We found no significant difference in cleaved caspase-3+ cell proportions between the groups (Control: 874 cells from 5 mice; Ghsr1a-KD: 678 cells from 7 mice), suggesting that ghrelin signaling does not influence the survival of new neurons and their progenitors. Similarly, the percentage of Ki67+ cells in the RMS was similar between the two groups (Figure S8), indicating that Ghsr1a-KD does not impair cell proliferation in the RMS. However, technical limitations prevented a reliable evaluation of proliferation in the V-SVZ,

as lentivirus injection into this region may interfere with GHSR1a expression in not only neural progenitors and new neurons, but also other GHSR1a-expressing cell types (Zigman et al., 2006). Although ghrelin signaling has been reported to promote cell proliferation in the V-SVZ (Li et al., 2014), our complementary in vitro KD experiments (Figure 4C–J) and in vivo OB-core lentivirus injections (Figure 5A–C), which did not affect the V-SVZ, consistently demonstrated that Ghnr1a-KD reduces neuronal migration. Taken together, our results suggest that blood-derived ghrelin enhances neuronal migration in the RMS and OB by stimulating actin cytoskeleton contraction in the cell soma, rather than by altering cell proliferation or survival.” (Results, Page 10, line 19–Page 11, line 4)

“rat anti-Ki67 (1:500, #14-5698-82, eBioscience); and rabbit anti-cleaved caspase-3 (1:200, #9661, Cell Signaling Technology)” (Methods, Page 48, lines 14–16)

How much is ghrelin/Ghsr1 signaling conserved in marmosets?

How ghrelin signaling is conserved between mice and common marmosets is important to clarify. A previous study reported the existence of a ghrelin homolog in common marmoset, which shares high sequence similarity with that in mice (Takemi et al., 2016). Moreover, the GHSR1a homolog in the common marmoset (<https://www.ncbi.nlm.nih.gov/protein/380748978>) shares 95.36% amino acid identity with its mouse counterpart. These findings suggest that blood-derived ghrelin may similarly promote neuronal migration in the marmoset brain, as observed in mice.

We have added the following text in the Discussion section:

“Our data showed that new neurons preferentially migrate along arteriole-side vessels rather than venule-side vessels in both mouse and common marmoset brains, suggesting that the mechanism of blood flow-dependent neuronal migration is conserved across rodent and primate species, as well as across brain regions. A previous study identified a ghrelin homolog in the common marmoset with high sequence similarity to the murine version (Takemi et al., 2016). In addition, the marmoset GHSR1a homolog shares 95.36% amino acid identity with that of the mouse (<https://www.ncbi.nlm.nih.gov/protein/380748978>). These findings suggest that blood-derived ghrelin promotes neuronal migration in the common marmoset brain in a manner similar to that in mice.” (Discussion, Page 15, lines 8–16)

Page 9. Starvation has been shown to boost ghrelin blood levels. What is the exact protocol used in this experiment and is this indeed increasing Ghrelin release from blood vessels in the V-SVZ? What about Ghsr1 expression level in newborn neurons?

We have clarified the calorie restriction (CR) protocol used in our experiments. We adopted a 70% CR protocol, which was previously shown to enhance hippocampal neurogenesis when administered for 14 days (Hornsby et al., 2016). In our study, the daily food intake under ad libitum (AL) conditions was first measured, and CR mice were then fed 70% of that amount for 5 consecutive days (see Figure 5I and Figure S10A).

To assess whether CR enhances ghrelin transcytosis into the brain parenchyma, we performed ELISA to quantify ghrelin levels in the OB and RMS. However, ghrelin concentrations were below the detection limit in both groups, precluding a direct comparison.

We also considered whether CR modulates the expression level of the ghrelin receptor GHSR1a. A recent study reported that fasting increased GHSR1a expression in the OB (Stark et al., 2024), raising the possibility that CR may exert a similar effect. To test this, we performed in situ hybridization and quantified Ghnr1a mRNA puncta in Dcx+ cells in the OB. No significant difference was found between the AL and CR groups (Figure S5B), suggesting that CR does not alter GHSR1a expression levels in new neurons.

Although we cannot exclude the possibility that CR increases GHSR1a expression in other OB cell types, our combined CR and Ghsl1a-KD experiments strongly support a cellautonomous contribution of ghrelin signaling to the enhanced neuronal migration observed under CR conditions. Corresponding data and text have been added to Figure S5 and the Results, Discussion, and the Figure legend sections as follows:

Minor

Page 4

Line 19 In Supplemental movies 1 and 2, it is unclear where to see the GFP+ new neurons interact with BV. Can you add arrows as an indication for the readers? It will be better to add the anatomy term for orientation, caudal, or rostral in the video. (The same for Supplemental movies 3, 4, and 5).

To clarify the regions of interest in Supplemental Movies 1 and 2, where neuron–vessel interactions in the RMS are highlighted, we added dotted lines indicating the RMS boundaries. In addition, we created a new movie (Supplemental Movie S1') showing a high-magnification view of Supplemental Movie S1, in which arrows mark EGFP+ new neurons interacting with blood vessels. We also added orientation indicators (e.g., caudal and rostral) and arrows to highlight new neuron–vessel interactions in Supplemental Movies S1–S5.

The following descriptions have been added to the Figure legends:

“Supplemental Movie S1'

High-magnification view extracted from Supplemental Movie S1. Arrows indicate EGFP+ cells interacting with blood vessels.” (Figure legend, Page 46, lines 6–8)

“Arrows indicate EGFP+ cells interacting with blood vessels.” (Figure legend, Supplemental Movie S3, Page 46, lines 16–17)

“Arrows indicate Dcx+ cells interacting with blood vessels.” (Figure legend, Supplemental Movies S4 and S5, Page 46, lines 21–22, 26–27)

Blood vessels are labeled in the Supplemental movies 2 and 3 by employing Flt1DsRed transgenic mice instead of RITC-Dex-GMA. However, Flt1-DsRed transgenic mice are not mentioned in the results section.

We have now included an explanation regarding the use of Flt1-DsRed mice, in which vascular endothelial cells were labeled with DsRed.

“To visualize blood vessels, we also used Flt1-DsRed transgenic mice, in which vascular endothelial cells were specifically labeled with DsRed (Matsumoto et al., 2012). Using DcxEGFP/Flt1-DsRed double transgenic mice, we observed close spatial relationships between new neurons and blood vessels (Supplemental Movies S2 and S3).” (Results, Page 4, lines 22–26)

Figure 5. Can you indicate (in the figure legend and the result section) the stage of the adult brain used for this experiment?

We used 6- to 12-week-old adult male mice in all experiments in this study. To specify this, we have added the age of animals to both the Results and the relevant Figure legends as follows:

“Therefore, we first studied blood vessel-guided neuronal migration in the RMS and OB using three-dimensional imaging in 6- to 12-week-old adult mice, which enabled analysis of the in

vivo spatial relationship between new neurons and blood vessels.” (Results, Page 4, lines 14–16)

“Figure 1 New neurons migrate along blood vessels with abundant flow in the adult brain.” (Figure legend, Page 25, line 4)

“(B, C) Three-dimensional reconstructed images of a new neuron (green) and blood vessels (red) in the rostral migratory stream (RMS) (B) and glomerular layer (GL) (C) of 6- to 12-week-old adult mice.” (Figure legend, Page 25, lines 6–8)

“(E) Transmission electron microscopy image of a new neuron (green) in close contact with a blood vessel (red) in the GL of a 6- to 12-week-old adult mouse.” (Figure legend, Page 26, lines 4–5)

“(F) Time-lapse images of a migrating neuron (indicated by asterisks) in the GL of a 6- to 12-week-old Dcx-EGFP mouse.” (Figure legend, Page 26, lines 6–7)

“Figure 3 Ghrelin is delivered from the bloodstream to the RMS and OB in the adult brain (A) Representative images of the OB and cortex of a fluorescent ghrelin-infused mouse (6 to 12 weeks old).” (Figure legend, Page 30, lines 1–3)

“Lentivirus injection into the OB core (A) and the VSVZ (D) was performed in 6- to 12-week-old adult mice.” (Figure legend, Page 33, lines 3–4)

Reviewer #2 (Recommendations for author):

Major:

*Ghsr1KD and blood flow 2-photon experiments to directly measure migratory speed.
Could also do the same with fasting with or without Ghsr1KD.*

We thank the reviewer for the valuable suggestion to strengthen our study. As pointed out in the Public Review, we agree that direct in vivo measurement of neuronal migration speed under Ghsr1a-KD conditions is important to clarify the link between ghrelin signaling and blood flow.

Two-photon imaging is the most suitable method for this purpose. Although we attempted two-photon imaging of Ghsr1a-KD new neurons, the number of virus-infected cells observed in vivo was too low to yield reliable data. Therefore, we chose an alternative strategy, combining Ghsr1a-KD with blood flow reduction using the BCAS model (Figure S9A), in which migration speed can be quantified based on the percentage of labeled cells reaching the OB. As stated in the Public Review response, BCAS significantly decreased the migration speed of Ghsr1a-KD new neurons (Figure S9B), indicating that Ghsr1a-KD does not abolish the influence of blood flow reduction. These findings suggest that ghrelin signaling is involved, but is not essential, for blood flow-dependent neuronal migration.

As suggested by the reviewer, direct observation of migration dynamics (e.g., somal translocation, leading process extension, stationary and migratory phases) is needed, especially in calorie restriction experiments. Although our data indicate that ghrelin signaling is required for fasting-induced increases in migration speed of new neurons, calorie restriction could also change concentrations of other factors in blood (Bonnet et al., 2020; Wu et al., 2024; Alogaiel et al., 2025), which may independently affect behavior of migrating neurons. Given that ghrelin is not the sole factor contributing to blood flow-dependent neuronal migration, other circulating factors could affect behavior of migrating neurons in a different manner during fasting. In vivo twophoton imaging would be a powerful approach to determine whether fasting-induced neuronal migration is caused by upregulated somal translocation speed, which would further support a role for ghrelin in this process.

We have added the following text in the Discussion:

“Although our data indicate that ghrelin signaling is essential for fasting-induced acceleration of neuronal migration, calorie restriction may also alter the concentrations of other circulating factors (Bonnet et al., 2020; Wu et al., 2024; Alogaiel et al., 2025), which could independently influence the behavior of migrating neurons.” (Discussion, Page 14, lines 25–29)

Minor:

(1) Show fluorescent Ghrelin in Figure 3 for all brain areas measured in Figure 1 (GL, EPL, GCL, and RMS) for direct comparison.

To allow for direct comparison across brain regions, we added a new Supplemental figure showing the distribution of fluorescently labeled ghrelin in the OB, including the GL, EPL, GCL and RMS. This comprehensive view highlights ghrelin localization relative to vasculature and migrating neurons in the regions analyzed in Figure 1.

(1) Figure 1, panel I is presented in a confusing manner. High blood flow points to 0 degrees, low blood flow to 180 degrees. It implies (unintentionally, I am sure) that low blood flow results in migration away from OB. Maybe plot separately?

We agree that the original presentation of Figure 1I could be misinterpreted as referring to anatomical orientation (i.e., toward or away from the OB). To avoid confusion, we revised the figure to categorize new neuron–vessel interactions into four groups according to (1) the angle between the migration direction and vessel axis (small or large), and (2) whether the new neuron is migrating toward or away from the direction of higher blood flow. This new presentation avoids implying a fixed anatomical direction and better reflects the relationship between local blood flow and neuronal migration behavior. The revised figure is presented as Supplemental Figure S1.

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